IZONNET WEB SOLUTION PVT. LTD.
CHIKHLI

An internship report submitted to

UKA TARSADIA UNIVERSITY

Final Year Student of
Bachelor of Technology
in
Information Technology

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2025

CERTIFICATE

This is to certify that work embodied in this report entitled Summer Internship as carried out **by Patel Kris Mukeshbhai (202203103510351)** in partial fulfilment of the degree of Bachelor of Technology in Information Technology, Chhotubhai Gopalbhai Patel Institute of Technology, UTU, Bardoli during the academic year 2025-2026.

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Certificate

This is to certify that Mr. **Kris Mukeshbhai Patel** of B.Tech (Information Technology) Roll No **202203103510351** has completed his Internship (26th May 2025 to 05th July 2025) on **.Net Developer** with Izonnet Web Solution Pvt. Ltd. during the academic year 2025-2026.

During the tenure mentioned above, Mr. Kris Mukeshbhai Patel has remained involved with his duties and responsibilities assigned to him. We found his competent and active with sincerity and determination.

Izonnet Web Solution Pvt. Ltd.
Director

Sincerely,
Nilesh Ahir

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Patel Kris Mukeshbhai (202203103510351)

ABOUT COMPANY

Vision :

- To be a leading Web Solutions company in the IT sector, progressing in our market position by delivering reliable, innovative, and user-friendly software services.

- We believe that our customers' growth is our growth, and commit to helping them achieve their business goals with accuracy and quality.

Mission :

- We aim to deliver top-quality, cost-effective IT solutions with a strong focus on customer satisfaction. By fostering long-term client relationships, embracing innovation, and exceeding expectations, we empower businesses worldwide to enhance productivity, efficiency, and success.

Products/Services :

- Izonnet Web Solution Pvt. Ltd. specializes in web and app development, providing IT services like e-commerce solutions, digital marketing, and particularly tailored travel sector solutions, including booking engines and API integrations.

Turnover :

- Achieved a turnover of ₹25–30 Lakhs+ .

Partners :

- There are 2 partners in Izonnet Web Solution Pvt. Ltd.

Customers :

- Diverse client base with domestic and international presence
- Long-term relationships driven by quality, trust, and customer satisfaction

Achievements :

- Successful delivery of multiple projects across industries
- Recognized expertise in travel sector IT solutions
- High client satisfaction and retention rates
- Adoption of innovative and emerging technologies
- Supportive work environment promoting employee growth
- Strong track record in cost-effective, high-quality IT solutions

ABOUT INTERNSHIP AREA

1. Application Development

- Web Application Development
  Developing the hospital management system using ASP.NET Core MVC with C#, HTML, CSS, JavaScript, and Entity Framework Core for database interaction with SQL Server.

  Modules are directly linked to your database tables:
  - Patients for patient records
  - Doctors and DoctorSchedules for medical staff and their availability
  - Billings for financial transactions
  - Rooms for bed allocation and availability
  - AdminUsers and Users for authentication and roles
  - ContactMessages for patient/staff queries.
- Cross-Platform Support
  Future-ready integration using Blazor or .NET MAUI to provide access via web, desktop, or mobile devices for hospital staff and patients.

2. UI/UX Design

- Building responsive dashboards for SuperAdmin, Admin, Doctors, and Receptionists.
- Creating clean forms for patient registration, billing entry, doctor scheduling, and room allocation.
- Styling with Bootstrap/TailwindCSS for consistent design and accessibility.

3. Backend Development

- Controller- Service- Repository Pattern for modular, maintainable code:
  - PatientsController → Patients table
  - DoctorsController → Doctors & DoctorSchedules tables
  - BillingController → Billings table
  - RoomsController → Rooms table
  - ContactController → ContactMessages table.
- RESTful APIs for external integration with mobile apps or third-party hospital systems.

4. Quality Assurance (QA) and Testing

- Unit Tests using xUnit/NUnit to verify CRUD operations for each table.
- Integration & UI Tests using Selenium to simulate real user workflows, such as booking an appointment or generating a bill.

5. App Maintenance and Updates

- Using EF Migrations (__EFMigrationsHistory table) to manage schema changes.
- Continuous improvement:
  - AI-based appointment scheduling from DoctorSchedules
  - Bed management automation from Rooms availability

6. Deployment and Distribution

- Hosting on IIS, Azure App Service, or on-premises servers.
- Automated deployment pipelines via Azure DevOps or GitHub Actions to ensure smooth updates.

7. Security and Compliance

- Role-based authentication via ASP.NET Identity:
  - AdminUsers for system admins
  - Users for doctors and other staff.
- Protecting patient records (Patients table) with encryption and access control.
- SQL injection prevention, data validation, and compliance with local healthcare laws.

8. Analytics and Performance Monitoring

- Using Application Insights/Serilog to monitor query performance and application health.
- Generating reports from:
  - Billings → Revenue trends
  - Rooms → Occupancy rates
  - DoctorSchedules → Doctor workload.

9. Project Management

- Following Agile with sprints mapped to database-driven modules:
  - Sprint 1 → Patients & Doctors
  - Sprint 2 → Billing & Rooms
  - Sprint 3 → Contact & Scheduling.
- Tracking progress in Jira/Trello/Azure Boards.

10. Client Consultation and Support

- Gathering requirements for custom modules like Lab Tests or Pharmacy.
- Managing queries via ContactMessages table.
- Offering staff training and 24/7 technical support post-deployment.

11. Emerging Technologies

- AI integration for predicting patient needs from Patients history.
- IoT integration for linking hospital devices to patient records.
- Machine Learning models for analyzing trends in Billings and treatment success rates.

LEARNING OUTPUT

Week 1 – Project Understanding & Requirement Gathering
In the first week of my internship, I focused on understanding the scope and objectives of the Hospital Management System (HMS).

I explored how hospitals digitally manage operations such as:

- **Patient registration** (Patients table)
- **Appointment booking** (DoctorSchedules & Patients)
- **Doctor scheduling** (DoctorSchedules table)
- **Room allocation** (Rooms table)
- **Billing** (Billings table)

I identified the different user roles - SuperAdmin, Admin, Doctor, and Patient and their respective functionalities.

I then prepared a roadmap for the modules, including Login/Register (Users, AdminUsers), Doctor Management, Patient Records, Room Management, and Billing.

This planning stage gave me a **clear vision** for the development process.

Week 2 – Database Design & Project Setup
In week two, I designed the database schema using SQL Server Management Studio 2021.

I created structured tables for core entities:

- Users & AdminUsers – authentication and role management
- Doctors – doctor details
- Patients – patient details and medical history
- DoctorSchedules – managing doctor availability
- Rooms – bed and ward allocation
- Billings – payments and invoices
- ContactMessages – user queries and feedback

I applied relationships and foreign keys to maintain data integrity.
At the same time, I set up the ASP.NET Core MVC project, configured the solution structure, and installed Entity Framework Core for ORM operations.

Week 3 – Core Backend Development
During the third week, I implemented core backend functionalities using Entity Framework Core:

- Created Models for database tables.
- Developed Controllers for managing Doctors and Patients with full CRUD operations.
- Built Razor Views to display and update dynamic data.

This week improved my understanding of MVC architecture, data binding, and server-side rendering in ASP.NET Core MVC.

Week 4 – UI/UX Development & Role-Based Access

In week four, I worked on the frontend UI using Razor Pages and Bootstrap for responsive and professional layouts.

- Improved form designs, tables, and navigation menus.
- Created a consistent layout structure across the application.
- Implemented Role-Based Access Control so that:
    - Admins can manage doctors, patients, rooms, and billing.
    - Doctors can view schedules and patient history.
    - Patients can only view their own records, bills, and appointments.

Week 5 – Advanced Features & Optimization

This week, I developed additional backend functionalities:

- Appointment Booking System linking Patients, Doctors, and DoctorSchedules.
- Billing Module with PDF generation using Rotativa.
- Applied Repository Pattern and ViewModels for a more maintainable architecture.
- Improved data validation and secure data handling.

This made the application more scalable, secure, and maintainable.

Week 6 – Testing, Bug Fixing & Documentation

In the final week, I focused on quality assurance and documentation:

- Conducted Unit Testing (xUnit) and manual testing to fix bugs.
- Standardized error handling, form validations, and UI consistency.
- Prepared project documentation including:
    - Database diagrams showing table relationships.
    - User flow charts for each role.
    - Screenshots of the major modules.

I also prepared my final internship report summarizing the modules, tools, and key learning outcomes.

This week gave me a complete understanding of the software lifecycle from planning to deployment.

TABLE OF CONTENTS

ACKNOWLEDGEMENT ..... **IV**
ABSTRACT ..... **V**
LIST OF FIGURES ..... **XI**
LIST OF TABLES ..... **XIII**
CHAPTER 1: INTRODUCTION **1**
1.1 Overview 1
1.2 Problem Definition 2
1.3 Scope/Application 2
CHAPTER 2: TRAINING ACTIVITIES **3**
CHAPTER 3: MODULES **4**
3.1 System Modules 4
3.1.1 User Management Module 5
3.1.2 Appointment Scheduling Module 5
3.2 Database Schema 6
CHAPTER 4: IMPLEMENTATION **10**
4.1 Hardware and Software Requirements 10
4.2 Snapshots 11
CONCLUSION AND FUTURE SCOPE **34**
REFERENCES **35**

LIST OF FIGURES

Figure 4.1 Login Page 11
Figure 4.2 Create User Account Page 11
Figure 4.3 Create Admin Account Page 12
Figure 4.4 Forgot Password Page 12
Figure 4.5 Home Page 13
Figure 4.6 Doctor Page 13
Figure 4.7 Add Doctor Page 14
Figure 4.8 View Doctor Page 15
Figure 4.9 Edit Doctor Page 16
Figure 4.10 Delete Doctor Page 17
Figure 4.11 Doctor Schedule Page 17
Figure 4.12 Add Doctor Schedule Page 18
Figure 4.13 View Doctor Schedule Page 18
Figure 4.14 Edit Doctor Schedule Page 19
Figure 4.15 Delete Doctor Schedule Page 19
Figure 4.16 Patients Page 20
Figure 4.17 Add Patients Page 20
Figure 4.18 View Patients Profile Page 21
Figure 4.19 Edit Patients Profile Page 22
Figure 4.20 Delete Patients Profile Page 23
Figure 4.21 Room Management Page 23
Figure 4.22 Add New Room Page 24
Figure 4.23 View Room Information Page 24
Figure 4.24 Edit Room Information Page 25
Figure 4.25 Delete Room Information Page 25
Figure 4.26 Billing Records Page 26
Figure 4.27 Create Billing Record Page 26
Figure 4.28 View Billing Summary Page 27
Figure 4.29 Edit Billing Record Page 28
Figure 4.30 Patient Billing Report Page 28
Figure 4.31 Admin User Page 29
Figure 4.32 Add Admin User Page 29

Figure 4.33 Edit Admin User Page 30
Figure 4.34 Delete Admin User Page 30
Figure 4.35 Hospital Report Summary page 31
Figure 4.36 About Page 31
Figure 4.37 Contact Page 32
Figure 4.38 Contact Messages Page 32
Figure 4.39 Gemini AI Assistant Page 33

LIST OF TABLES

Table 3.1 Users 6
Table 3.2 Doctors 6
Table 3.3 DoctorsSchedule 7
Table 3.4 Patient 7
Table 3.5 Room 8
Table 3.6 Billing 8
Table 3.7 AdminUser 9
Table 3.8 ContactMessage 9

CHAPTER 1: INTRODUCTION

1.1 OVERVIEW

Overview of the Hospital Management System Project

The Hospital Management System (HMS) project is designed to automate and streamline the core administrative, clinical, and financial operations of a hospital. Built using ASP.NET Core MVC and SQL Server, this system provides a centralized platform for managing doctors, patients, appointments, room allocations, billing, and role-based access.

Purpose of Initiation

- **Operational Efficiency:** To reduce manual paperwork and errors by digitizing hospital operations such as patient registration, doctor scheduling, room management, and billing.
- **Improved Patient Care:** Enabling faster, more accurate, and accessible patient information helps doctors and medical staff deliver better care.
- **Secure and Scalable Architecture:** The system is designed with modern web development technologies (.NET Core, Entity Framework) to ensure performance, security, and future scalability.

Prerequisites

- **Technical Expertise:** Knowledge of ASP.NET Core MVC, Entity Framework Core, SQL Server, HTML/CSS/Bootstrap, and Razor Pages is essential for system development.
- **Infrastructure:** Development and deployment environments, access to sample medical data, and appropriate testing tools for functional and security testing.
- **Stakeholder Input:** Coordination with hospital administrators, doctors, and staff to collect functional requirements and validate system workflows.

Expected Results

- **Centralized Management:** A unified system for handling patient data, appointments, doctor schedules, billing, and reports.
- **Secure Role-Based Access:** Access control based on roles (Admin, Doctor, Patient) ensuring data privacy and operational clarity.
- **Enhanced Reporting:** Automated billing and PDF report generation, appointment tracking, and patient history access.
- **Improved User Experience:** A user-friendly and intuitive interface tailored for all hospital staff and patients.

1.2 PROBLEM DEFINITION

Problem Definition for Hospital Management System Project

- **Manual Process Inefficiencies:** Many hospitals still rely on paper-based or semi-digital processes, which are time-consuming, error-prone, and difficult to scale.
- **Data Fragmentation:** Patient information, doctor schedules, and billing records are often stored separately, making retrieval and synchronization difficult.
- **Lack of Real-Time Access:** Doctors and patients often face delays in accessing medical history or appointment status due to the absence of a centralized system.
- **Security and Privacy Risks:** Without a role-based digital system, sensitive patient data can be exposed or mismanaged.
- **Scalability Limitations:** Legacy systems lack the flexibility to expand with hospital growth, new departments, or technology integrations.

1.3 SCOPE / APPLICATION

Scope of the Hospital Management System

The HMS project covers the digitization of various departments and workflows of a hospital. The system includes the following modules:

- User Authentication & Role-Based Access: For Admin, Doctor, and Patient logins.
- Doctor Management: Add, edit, delete, and view doctor profiles and schedules.
- Patient Management: Register new patients, maintain records, and track treatment history.
- Appointment Scheduling: Allow patients to book and manage doctor appointments.
- Room Allocation: Manage room availability and patient room assignments.
- Billing System: Generate and print medical bills, with options for discounts and status tracking.
- Contact Module: Allows users to submit inquiries or feedback.
- Reports: Generate patient history, bills, and other relevant reports in PDF format.

Application of the Hospital Management System

- Private and Public Hospitals: The system can be deployed in hospitals of any size for managing patient data, appointments, and billing efficiently.
- Clinics and Specialty Centers: Useful for small and medium-sized healthcare facilities to manage operations with minimal manual intervention.
- Telemedicine Platforms: With minor extensions, the system can be adapted for online consultation, appointment scheduling, and e-billing.
- Educational Institutions and Training: Hospitals connected to medical colleges can use the system for record-keeping, demonstration, and training purposes.

CHAPTER 2: TRAINING ACTIVITIES

ASP.NET Core is a modern, cross-platform framework developed by Microsoft for building dynamic web applications and APIs. It provides a robust environment for creating scalable, maintainable, and secure web solutions. In this project, ASP.NET Core MVC was used to develop a full-featured Hospital Management System, integrating both frontend and backend components with SQL Server as the database.

Key Components & Skills:

- C# Programming Language:
  Gained in-depth experience with C# for backend development, including object-oriented programming (OOP), asynchronous operations, and data validation.
- ASP.NET Core MVC Architecture:
  Used the Model-View-Controller pattern to separate concerns, allowing for clean code organization and easier maintenance. Implemented controllers, models, and Razor views to handle data flow and UI rendering.
- Entity Framework Core:
  Utilized EF Core as the Object-Relational Mapping (ORM) tool to perform CRUD operations and manage database interactions through code-first migrations and LINQ queries.
- Razor Pages and Layout Design:
  Designed user-friendly and responsive interfaces using Razor Pages combined with HTML, CSS, and Bootstrap. Created consistent layout templates for all roles, including Admin, Doctor, and Patient.
- Role-Based Access Control:
  Implemented secure login, registration, and authorization using ASP.NET Identity. Defined roles such as SuperAdmin, Admin, Doctor, and Patient to restrict access based on user privileges.
- SQL Server Integration:
  Created and managed a normalized database schema in SQL Server Management Studio. Included key entities such as Users, Doctors, Patients, Appointments, and Billings, with foreign key relationships for data integrity.
- RESTful API Development (Optional/Future Scope):
  Gained an understanding of how to build APIs for future mobile or third-party integration using ASP.NET Core Web API.
- Testing and Debugging:
  Debugged the application using Visual Studio tools and wrote unit tests with xUnit to ensure the stability and correctness of core modules like patient management and billing.
- PDF Generation and Reporting:
  Integrated tools like Rotativa to export appointment summaries and billing invoices as downloadable PDF reports for printing or email delivery.

Application: These skills were applied in the development of a full-stack hospital management system, enabling modules like patient registration, doctor scheduling, appointment booking, and billing. The application supports secure role-based access, responsive design for better user experience, and real-time data handling through SQL Server. This project strengthened my ability to build enterprise-level web applications aligned with industry standards.

CHAPTER 3: MODULES

3.1 SYSTEM MODULES

System Modules in the Hospital Management System

1. **User Management Module**
   - **Role**: Manages registration, authentication, and access control for different users: Admin, Doctor, and Patient.
   - **Function**: Ensures secure login, profile management, and differentiated access to hospital functionalities based on user roles.
2. **Patient Management Module**
   - **Role**: Handles the registration, tracking, and history of patients.
   - **Function**: Stores patient data, allows updates, and maintains complete medical records and visit history.
3. **Doctor Management Module**
   - **Role**: Manages doctor profiles, their availability, and specializations.
   - **Function**: Allows admins to add/edit/remove doctors, and lets doctors manage their schedules and view patient appointments.
4. **Appointment Scheduling Module**
   - **Role**: Facilitates appointment bookings between patients and doctors.
   - **Function**: Patients can view available time slots, request appointments, and receive confirmation, while doctors manage their upcoming appointments.
5. **Room Allocation Module**
   - **Role**: Tracks room availability and patient room assignments.
   - **Function**: Helps administrators assign and manage rooms for admitted patients, avoiding double bookings.
6. **Billing and Invoice Module**
   - **Role**: Generates and manages billing information based on treatments, consultations, room charges, and additional services.
   - **Function**: Automatically calculates total charges, applies discounts, generates invoices, and allows printing/export to PDF.
7. **Report Generation Module**
   - **Role**: Produces essential hospital reports such as patient records, bills, and appointment summaries.
   - **Function**: Exports structured reports in downloadable formats (PDF) for audits, management reviews, or patient handovers.
8. **Contact/Feedback Module**
   - **Role**: Enables users to contact hospital staff or leave feedback.
   - **Function**: Provides a form for submitting messages, which are stored and managed by admin for further action.

3.1.1 MODULE 1: User Management Module

Role and Functions:

1. **User Registration and Authentication**
   - Patients can register and log in using email, username, and password.
   - Admin and Doctors are added by Admin with role-based credentials.
   - Secure login using password hashing; supports session-based authentication.

2. **Role-Based Access Control (RBAC)**
   - Ensures:
     - **Admin** has access to all modules (Add/Remove Doctors, View Appointments, Manage Billing, Rooms, etc.).
     - **Doctors** can view assigned patients, manage schedules, and update medical notes.
     - **Patients** can request appointments, view status, and access bills.
3. **Profile Management**
   - Users (Doctors and Patients) can view and edit personal details.
   - Admin can update roles and profiles from the dashboard.
4. **Session and Security Management**
   - Token/session-based login system for secured access.
   - Includes logout and automatic session expiration features.
5. **Account Recovery**
   - "Forgot Password" functionality using email-based password reset.
   - Password strength validation and change password options available.

3.1.2 MODULE 2: Appointment Scheduling Module

Role and Functions:

1. **Appointment Booking (Patient Side)**
   - Patients can:
     - View doctor availability by department or name.
     - Select available time slots and book appointments.
     - Receive status updates (pending, approved, rejected).
2. **Appointment Management (Admin and Doctor Side)**
   - **Admins** can:
     - View all appointments.
     - Approve, reschedule, or reject requests.
   - **Doctors** can:
     - View upcoming appointments.
     - Add treatment or diagnosis notes after consultations.
3. **Notification System**
   - Displays status messages on dashboard (e.g., "Appointment Approved").
   - Email or in-app alert system can be integrated (optional enhancement).
4. **Patient Appointment History**
   - Patients can track their past and upcoming appointments.
   - Doctors can access visit history during consultations for better treatment decisions.
5. **Time Slot and Conflict Management**
   - Prevents double booking of doctors by disabling unavailable time slots.
   - Doctors can update availability, and system dynamically updates schedule.

3.2 DATABASE SCHEMA

The Hospital Management System project uses a SQL Server database to store and manage data efficiently. All the main tables—such as patients, doctors, users, rooms, billing, and feedback - were created and managed using SQL Server Management Studio.

Table 3.1 Users

Column Name	Data Type	Size	Constraint	Description
Id	INT	5	Primary Key	ID No.
Username	VARCHAR	50	---	Username
Password	VARCHAR	50	---	Password
Role	VARCHAR	20	---	User Role

Table 3.2 Doctors

Column Name	Data Type	Size	Constraint	Description
Id	INT	5	Primary Key	ID No.
Name	VARCHAR	100	---	Doctor's Full Name
Specialty	VARCHAR	50	---	Doctor's Specialization
Phone	VARCHAR	15	---	Contact Number
Email	VARCHAR	100	---	Email Address
Education	VARCHAR	50	---	Degree or Qualification
Address	VARCHAR	150	---	Residential or Clinic Address
About	TEXT	---	---	Short Bio or Introduction
ImagePath	VARCHAR	255	---	Path of Stored Profile Image

Table 3.3 DoctorSchedule

Column Name	Data Type	Size	Constraint	Description
Id	INT	---	Primary Key	Unique Schedule ID
DoctorId	INT	---	Foreign Key	Linked Doctor's ID
DayOfWeek	VARCHAR	20	---	Day of the Week
StartTime	TIME	---	---	Schedule Start Time
EndTime	TIME	---	---	Schedule End Time

Table 3.4 Patients

Column Name	Data Type	Size	Constraint	Description
Id	INT	5	Primary Key	Patient ID No.
FullName	VARCHAR	100	---	Patient's Full Name
Age	VARCHAR	3	---	Age of the Patient
Gender	VARCHAR	10	---	Gender (Male/Female/Other)
ContactNumber	VARCHAR	15	---	Patient's Mobile Number
Disease	VARCHAR	100	---	Disease/Illness
AdmissionDate	DATETIME	---	---	Date of Admission
Address	VARCHAR	150	---	Patient's Residential Address
ImagePath	VARCHAR	255	---	Profile Image Path
DiseaseDescription	TEXT	---	---	Summary of Patient's Condition

Table 3.5 Rooms

Column Name	Data Type	Size	Constraint	Description
Id	INT	5	Primary Key	Unique Room Entry ID
RoomNumber	INT	5	---	Room Number in the Hospital
Type	VARCHAR	20	---	Type of Room (ICU, Private, General)
Status	VARCHAR	20	---	Room Status (Occupied/Available)
PatientId	INT	5	Foreign Key (nullable)	ID of the Patient Assigned to the Room

Table 3.6 Billing

Column Name	Data Type	Size	Constraint	Description
Id	INT	5	Primary Key	Unique Billing Record ID
PatientName	VARCHAR	100	---	Name of the Patient
Amount	DECIMAL	10,2	---	Total Billing Amount
BillingDate	DATETIME	---	---	Date and Time of Billing
Description	TEXT	---	---	Details of Treatment or Billing Explanation

Table 3.7 AdminUsers

Column Name	Data Type	Size	Constraint	Description
Id	INT	5	Primary Key	Unique User ID
Username	VARCHAR	100	---	Name of the User
Password	VARCHAR	100	---	User's Login Password
Role	VARCHAR	20	---	User Role (User/Admin/SuperAdmin)

Table 3.8 ContactMessages

Column Name	Data Type	Size	Constraint	Description
Id	INT	5	Primary Key	Unique Feedback ID
Name	VARCHAR	100	---	Name of the person giving feedback
Email	VARCHAR	100	---	Email Address
Message	TEXT	---	---	Feedback or Comment
Date	DATETIME	---	---	Date and Time of Feedback

CHAPTER 4: IMPLEMENTATION

4.1 HARDWARE AND SOFTWARE REQUIREMENTS

HARDWARE REQUIREMENTS

1. **Minimum Requirements**
   - **CPU**: Dual-core 64-bit processor (Intel Core i3 / AMD Ryzen 3 or equivalent)
   - **RAM**: Minimum 4 GB
   - **Disk Space**: Minimum 5 GB of available disk space for Visual Studio, SQL Server, and project files
   - **Display**: 1366 x 768 screen resolution
   - **Network**: Basic internet access for installing dependencies and updates
2. **Recommended Requirements**
   - **CPU**: Quad-core 64-bit processor (Intel Core i5/i7 or AMD Ryzen 5/7)
   - **RAM**: 8 GB or more (16 GB recommended for smoother development experience)
   - **Disk Space**: SSD with at least 20 GB of free space for faster read/write operations
   - **Display**: Full HD (1920 x 1080) resolution for better UI/UX development
   - **Other**: UPS (Uninterrupted Power Supply) to prevent data loss during power outages

SOFTWARE REQUIREMENTS

1. **Operating System**
   - **Windows**: Windows 10 (64-bit) or later (Windows 11 recommended)
   - **Others (for reference only)**: While .NET Core supports cross-platform development, this project is optimized for Windows OS due to Visual Studio and SQL Server usage.
2. **Development Tools**
   - **IDE**: Visual Studio 2022 or later (Community/Professional/Enterprise Edition)
   - **.NET SDK**: .NET 6.0 or .NET 7.0 SDK (based on the version used)
   - **Database**: SQL Server 2019 or later (SQL Server Management Studio 18 or later for management)
   - **Entity Framework Core**: Installed as NuGet packages in the project for ORM (Object-Relational Mapping)
   - **Package Manager**: NuGet Package Manager (integrated in Visual Studio)
3. **Web Technologies**
   - **ASP.NET Core MVC**: Framework for building the web application
   - **HTML, CSS, JavaScript**: For front-end development
   - **Bootstrap 5**: For responsive and user-friendly UI
   - **Razor Pages**: For dynamic rendering and server-side logic
   - **Rotativa.AspNetCore**: For generating PDF reports from views
4. **Additional Tools and Frameworks**
   - **Git**: For version control (GitHub/GitLab/Bitbucket integration recommended)
   - **Microsoft Edge / Chrome**: For browser-based testing
   - **Postman** (optional): For testing API endpoints (if any)
   - **Microsoft Office / PDF Reader**: For viewing project documentation and reports

4.2 SNAPSHOTS

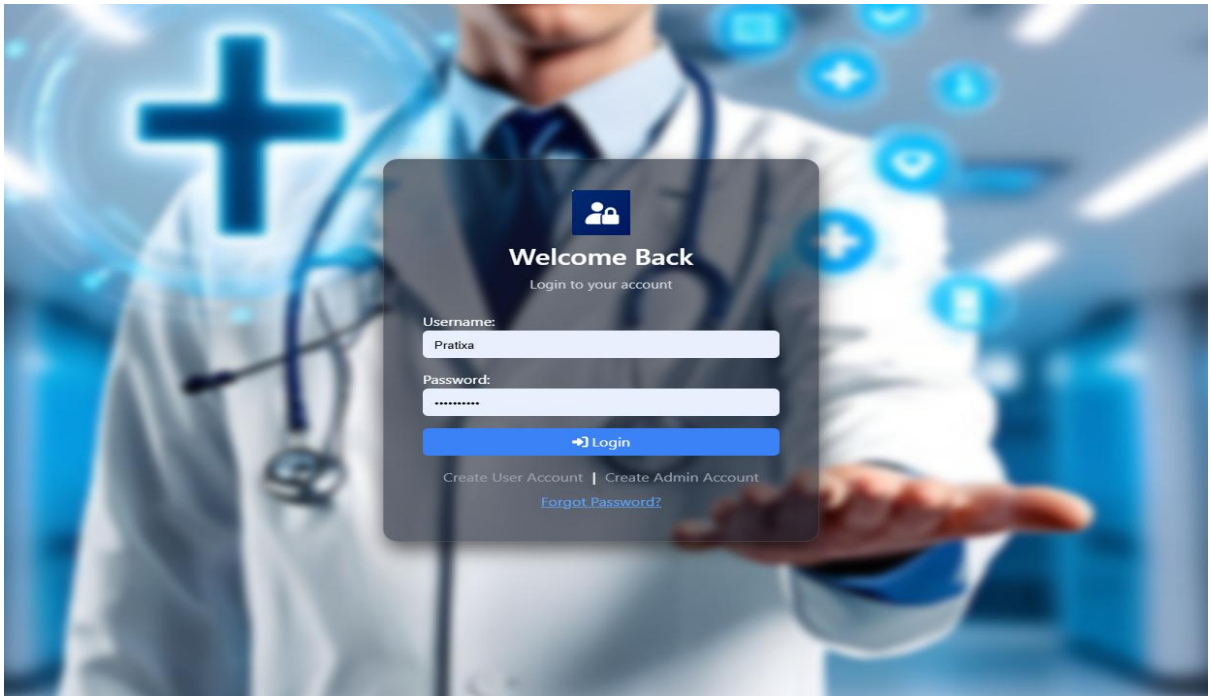


Figure 4.1 Login Page

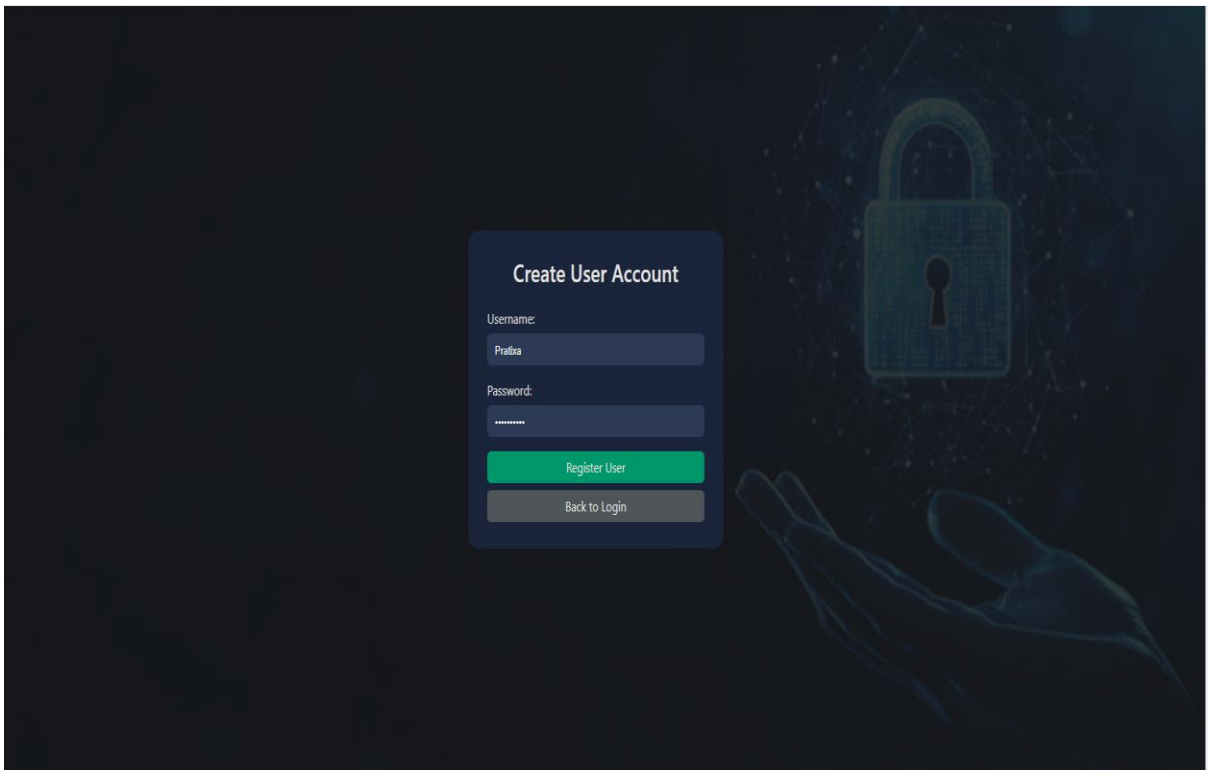


Figure 4.2 Create User Account Page

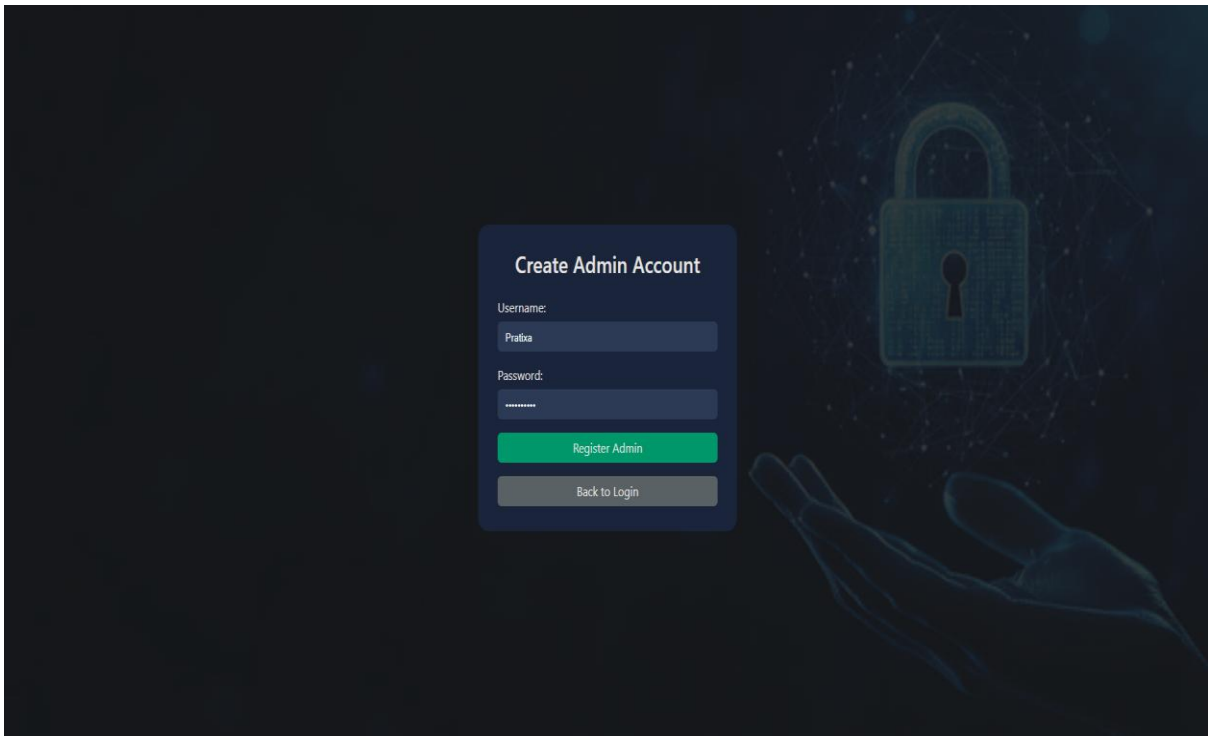


Figure 4.3 Create Admin Account Page

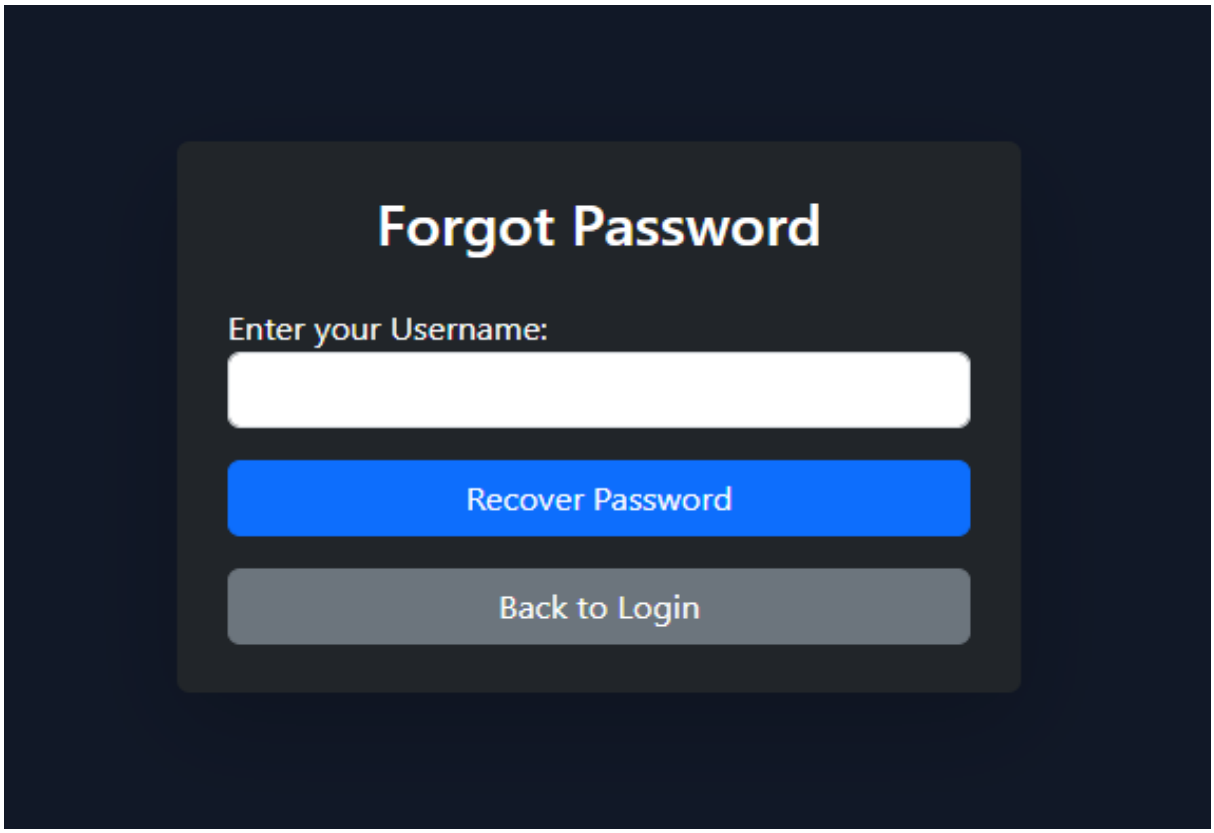


Figure 4.4 Forgot Password Page

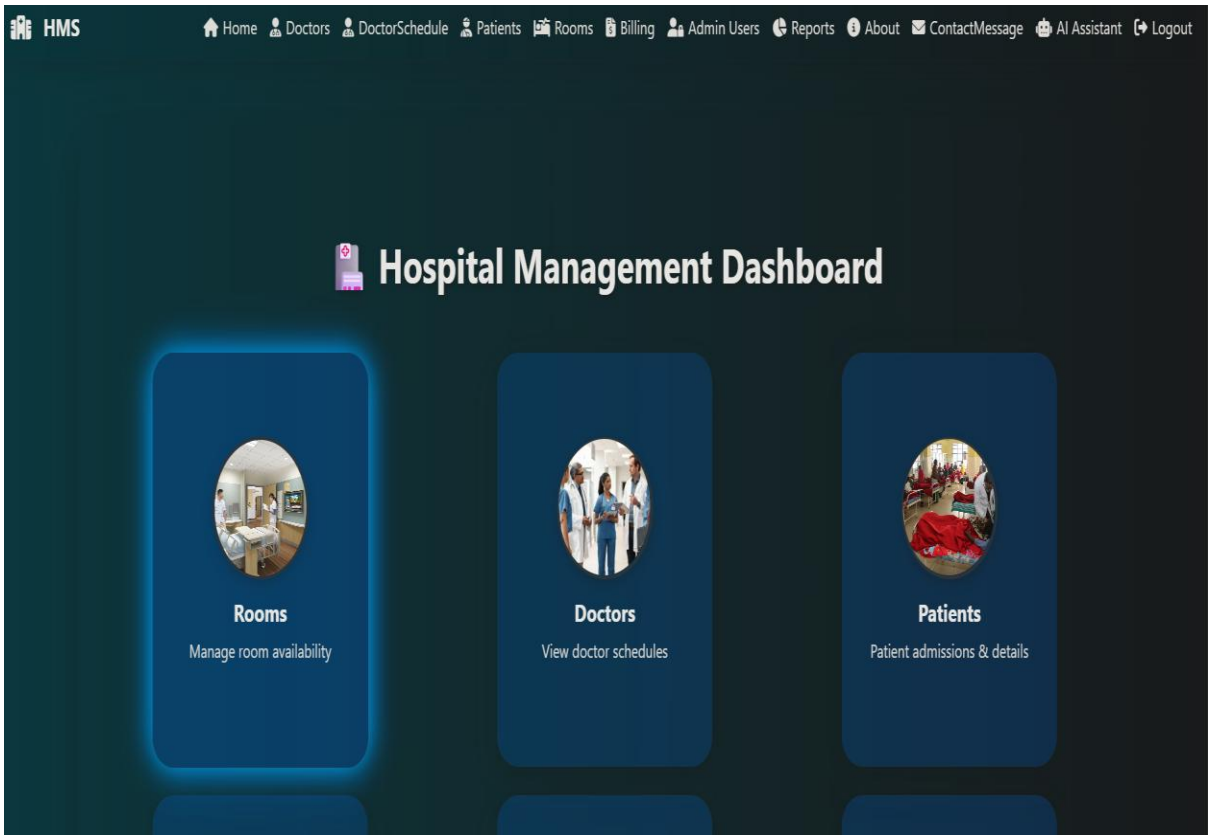


Figure 4.5 Home Page

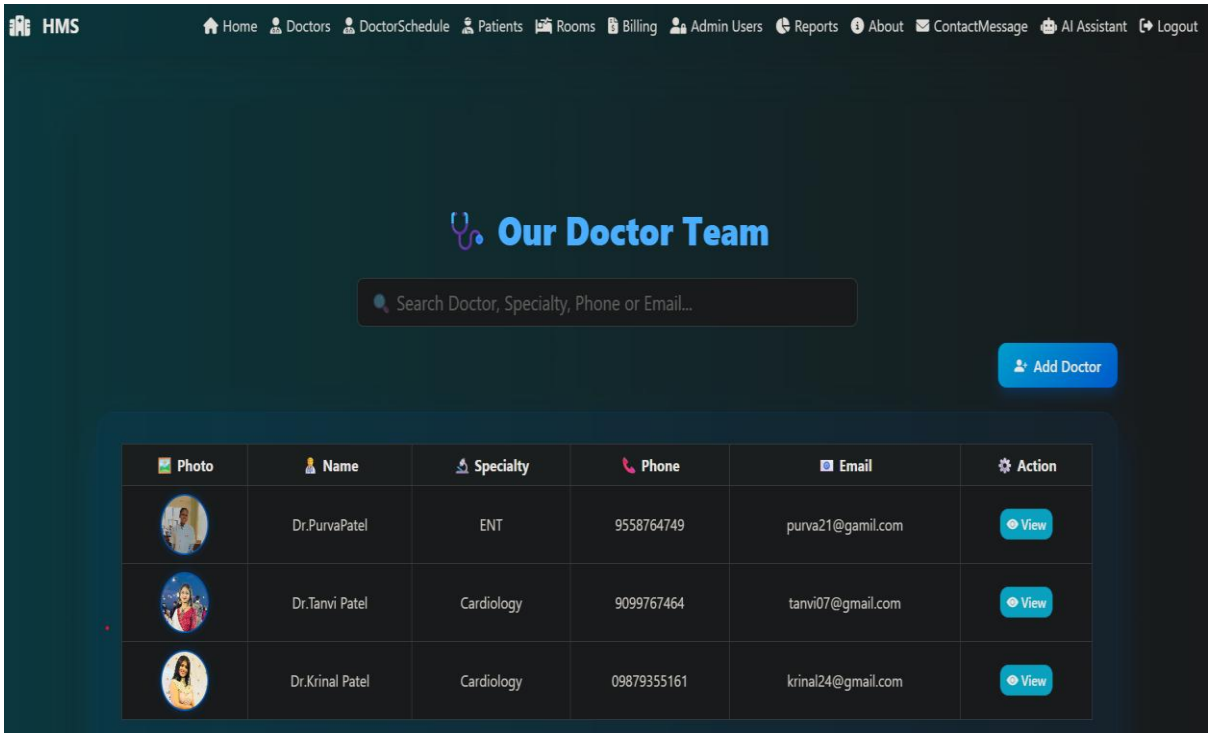


Figure 4.6 Doctor Page

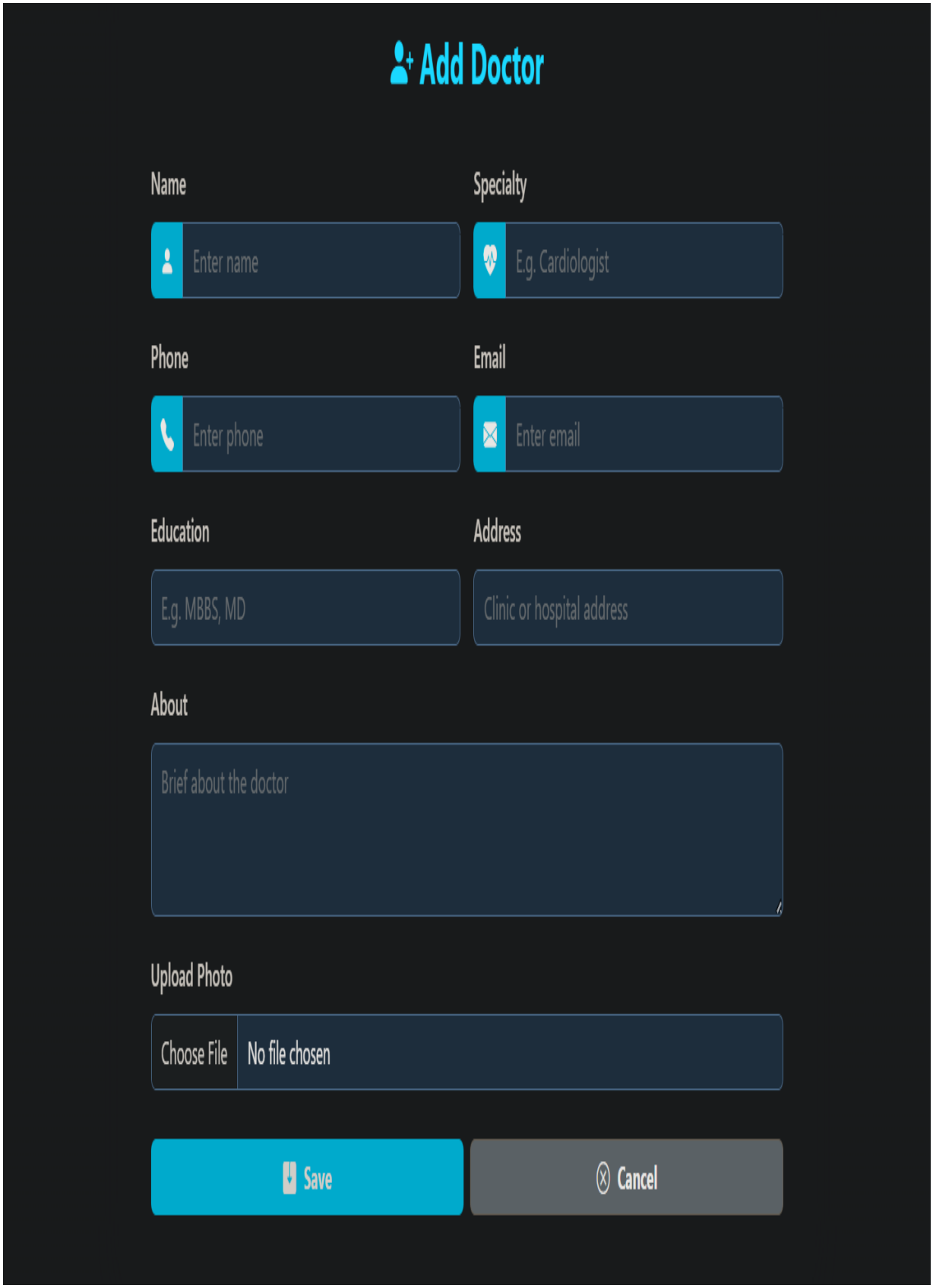


Figure 4.7 Add Doctor Page

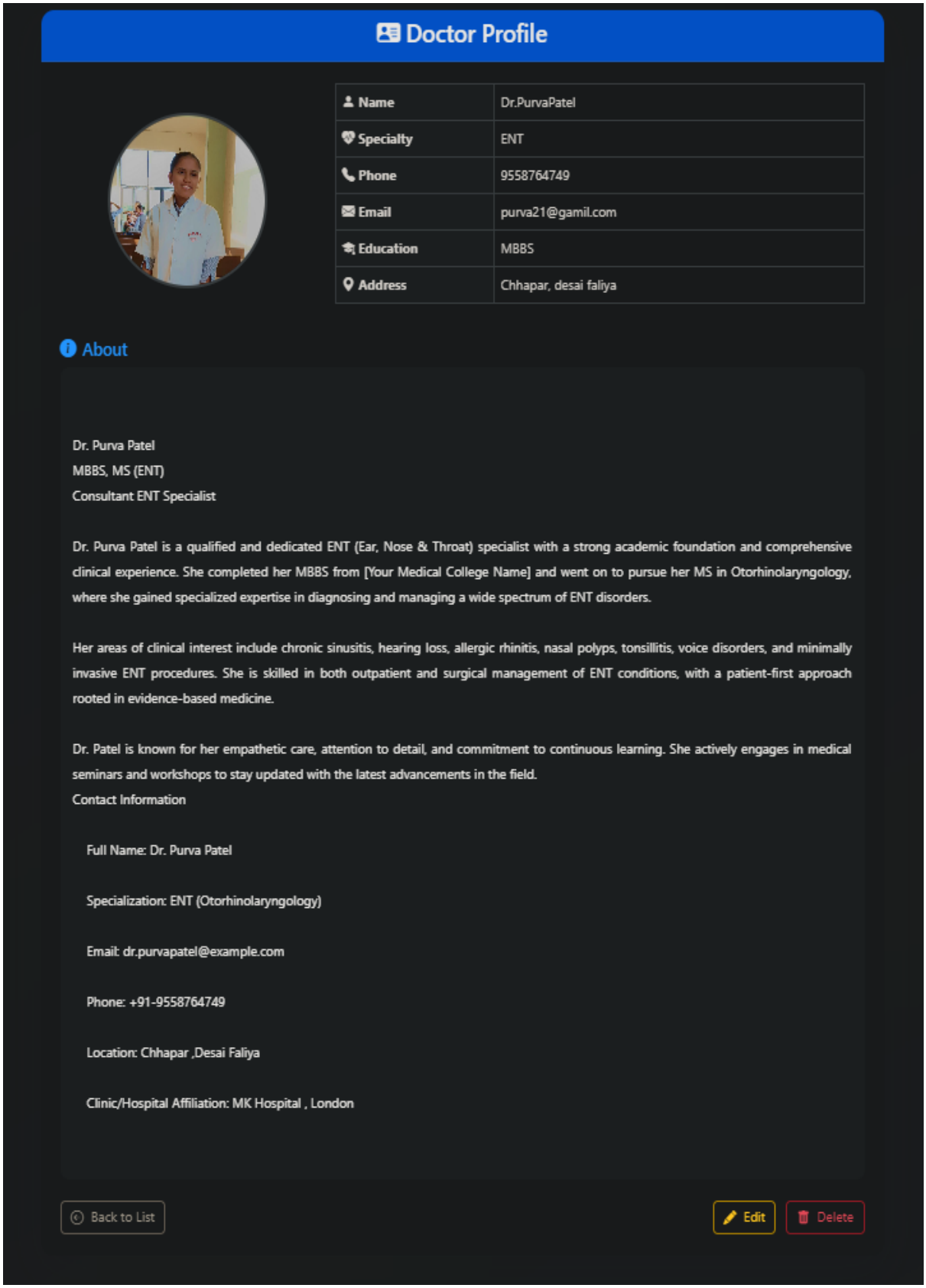


Figure 4.8 View Doctor Page

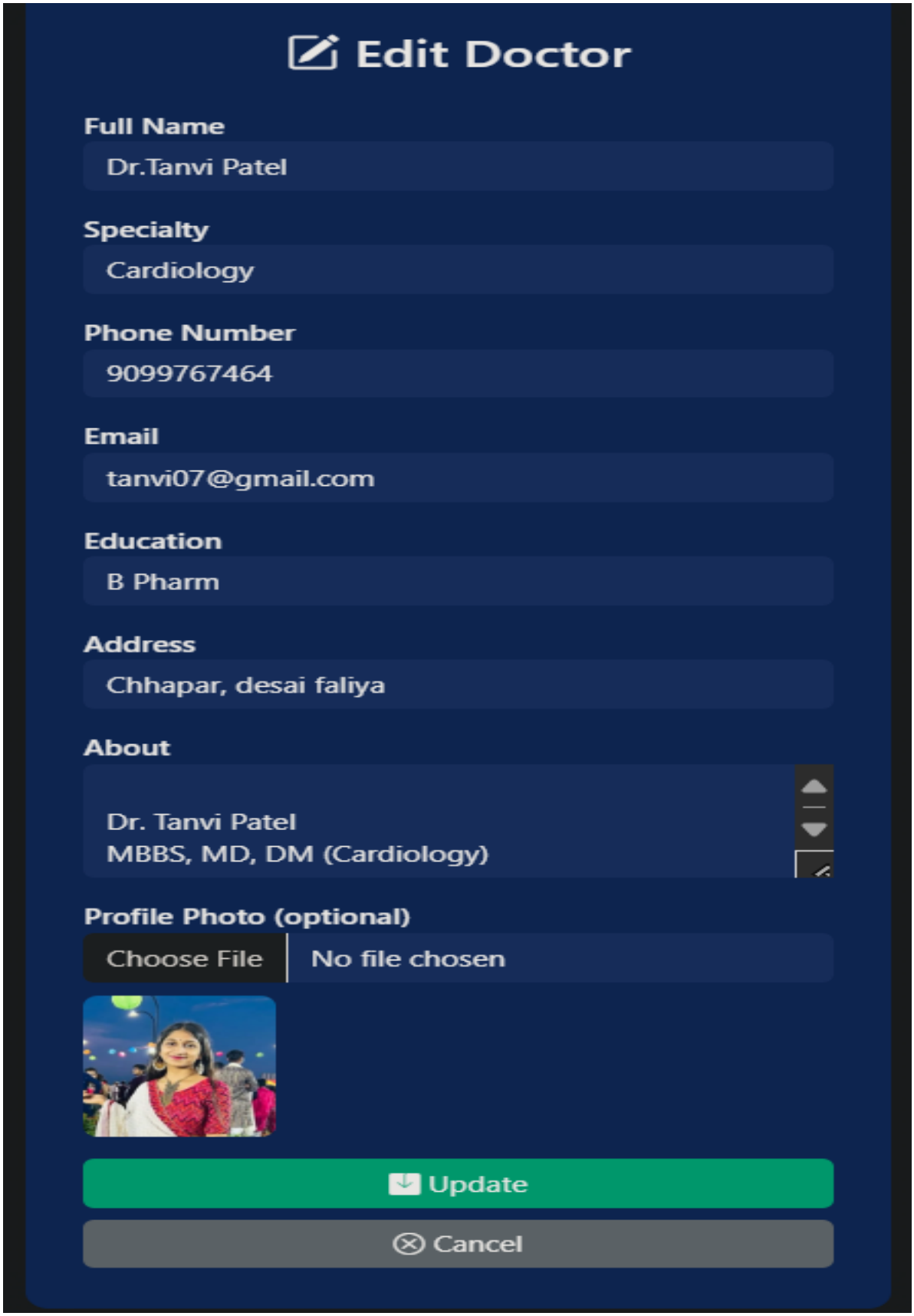


Figure 4.9 Edit Doctor Page

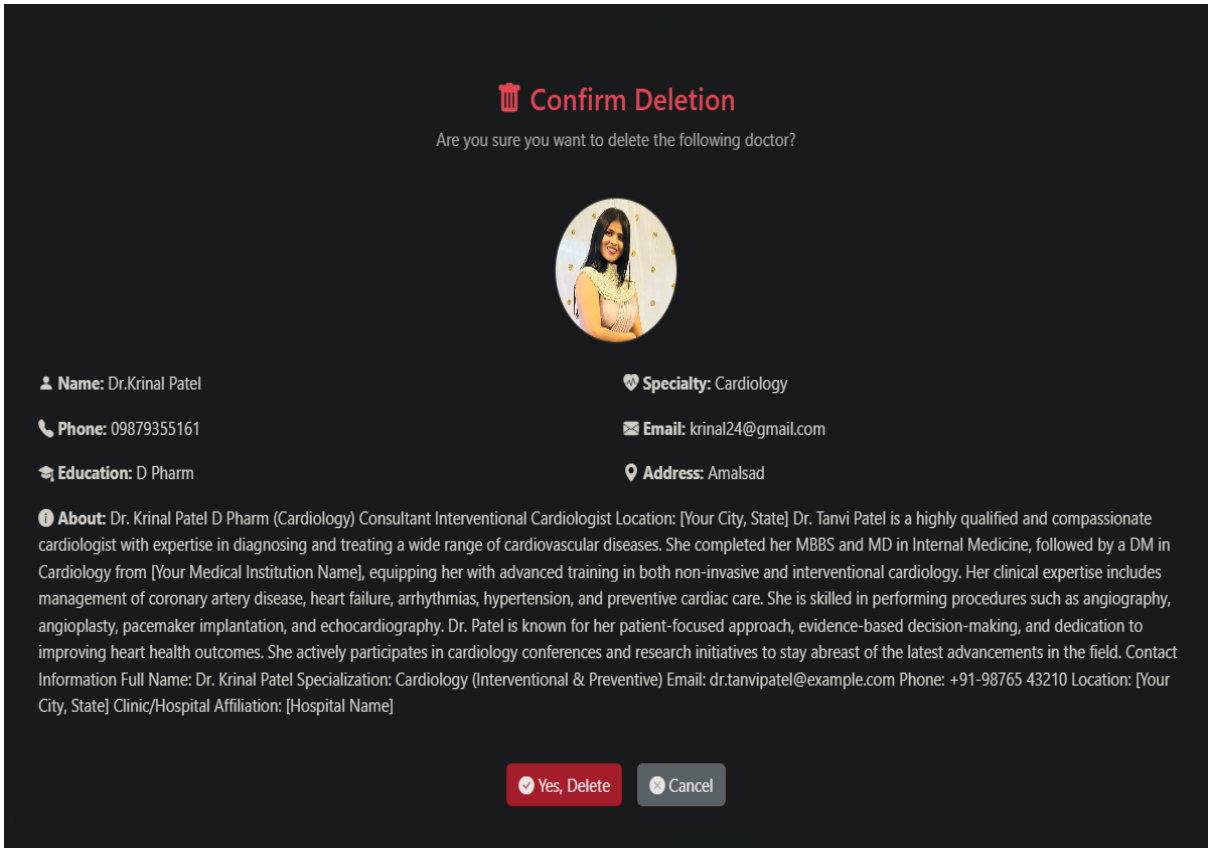


Figure 4.10 Delete Doctor Page

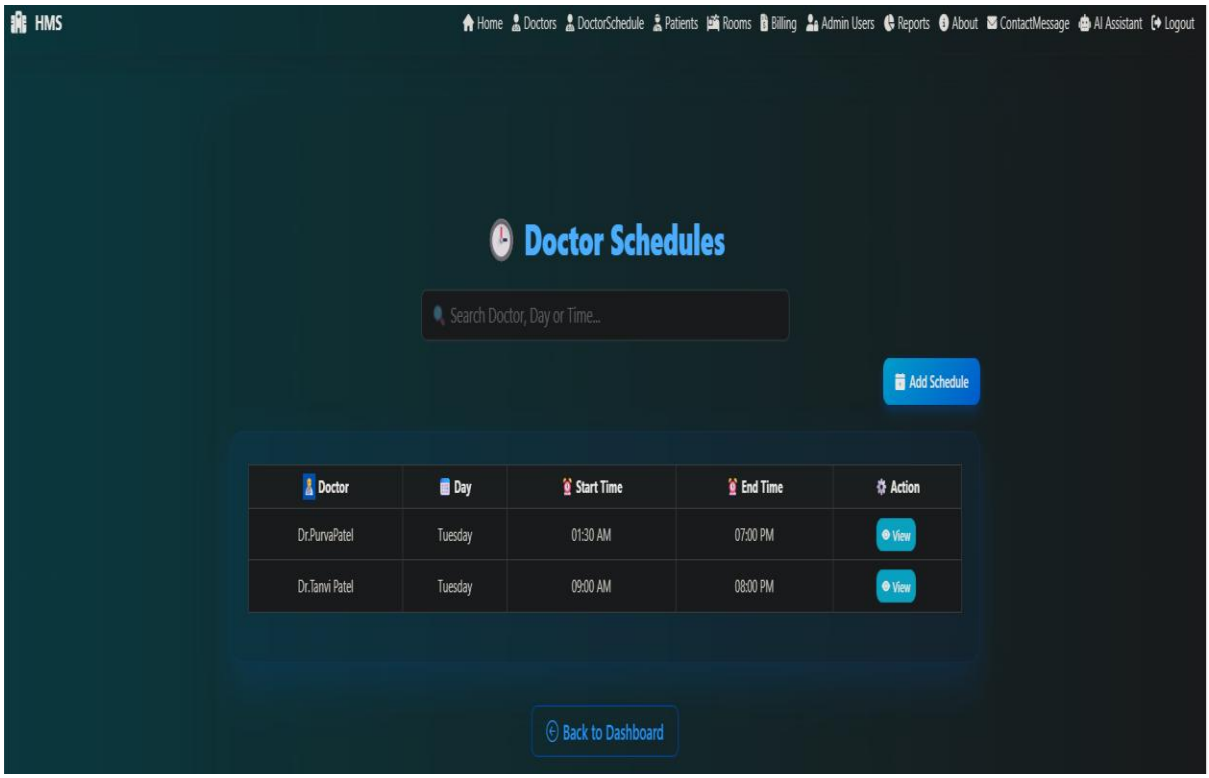


Figure 4.11 Doctor Schedule Page

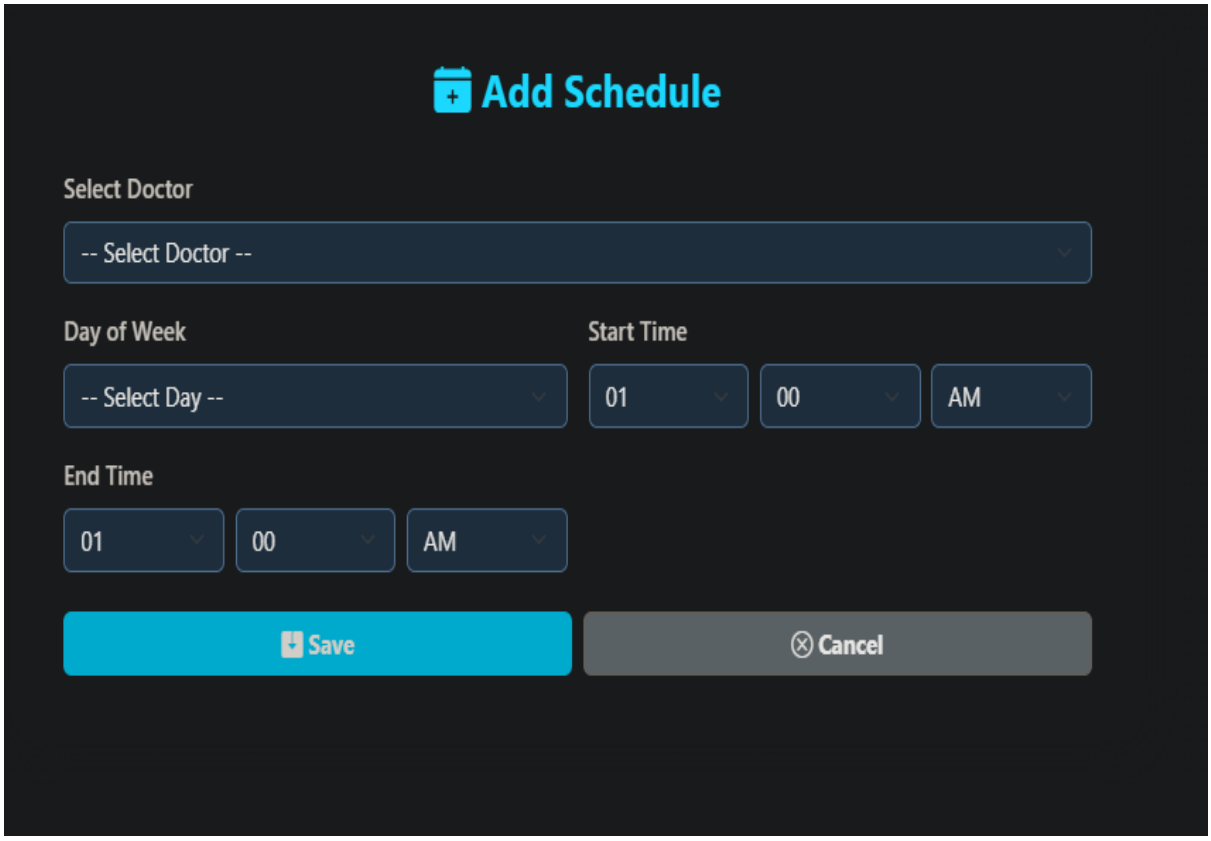


Figure 4.12 Add Doctor Schedule Page

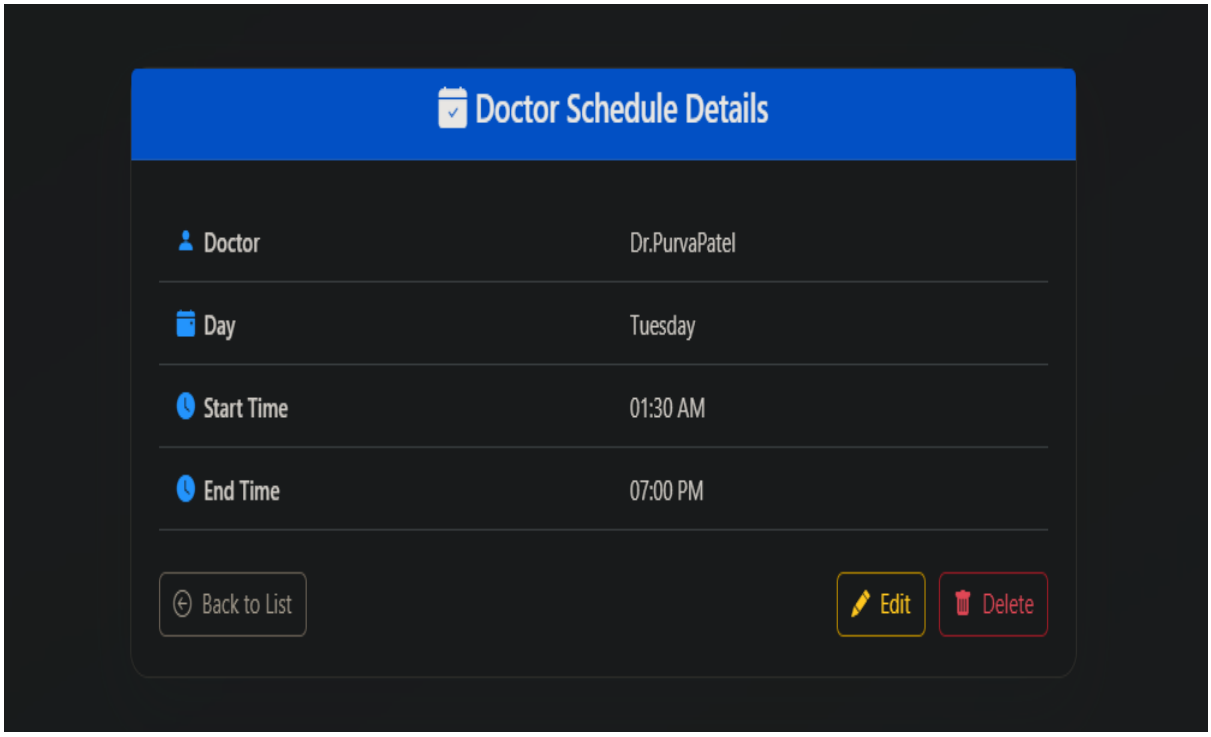


Figure 4.13 View Doctor Schedule Page

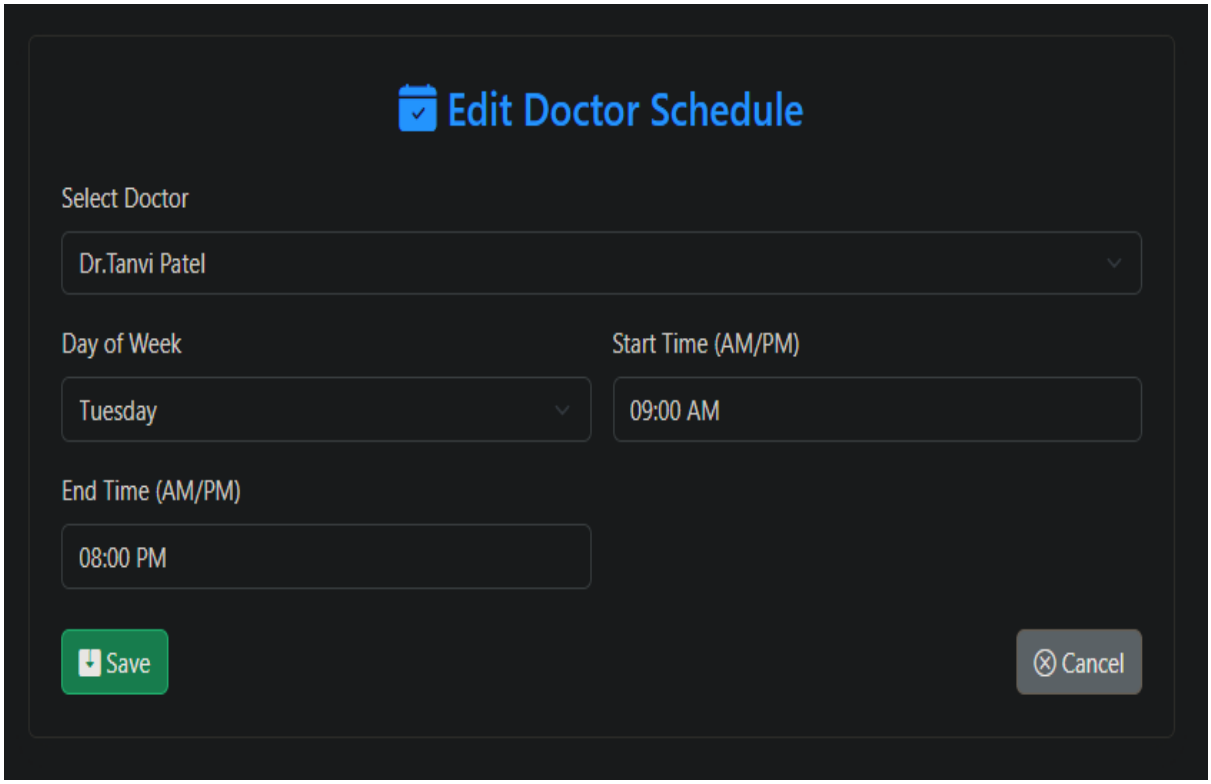


Figure 4.14 Edit Doctor Schedule Page

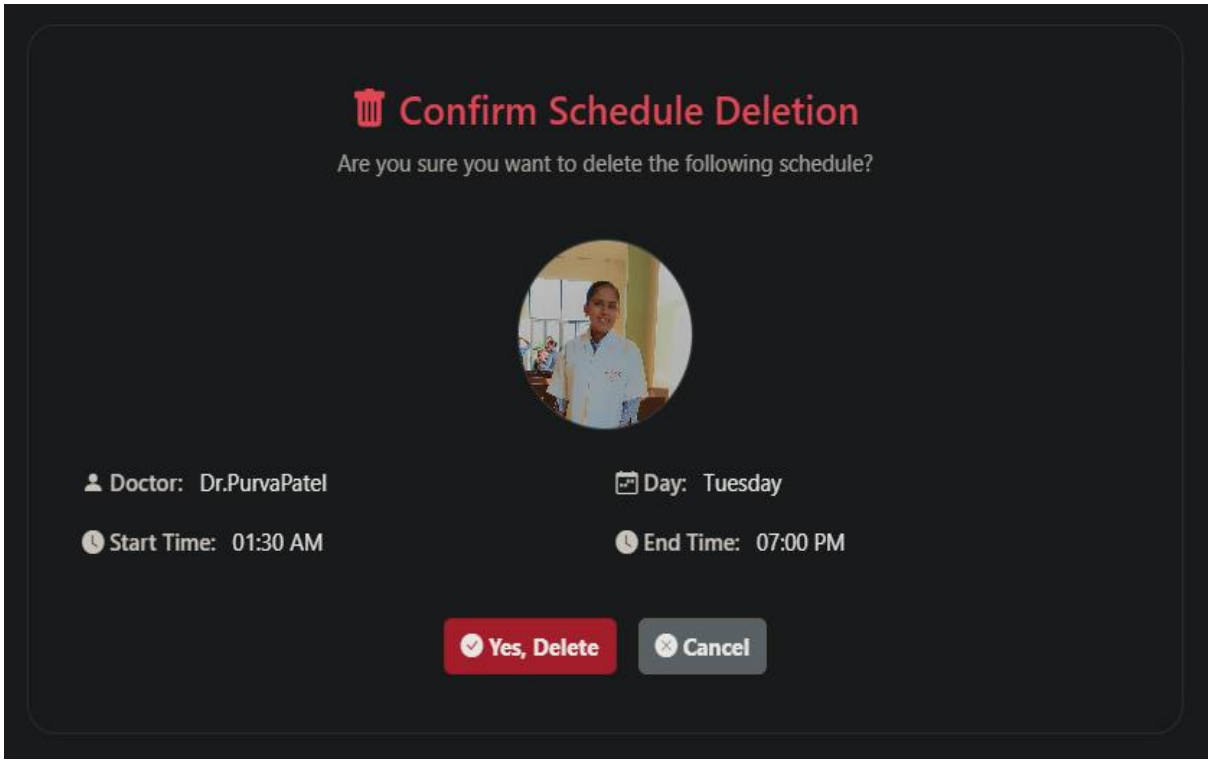


Figure 4.15 Delete Doctor Schedule Page

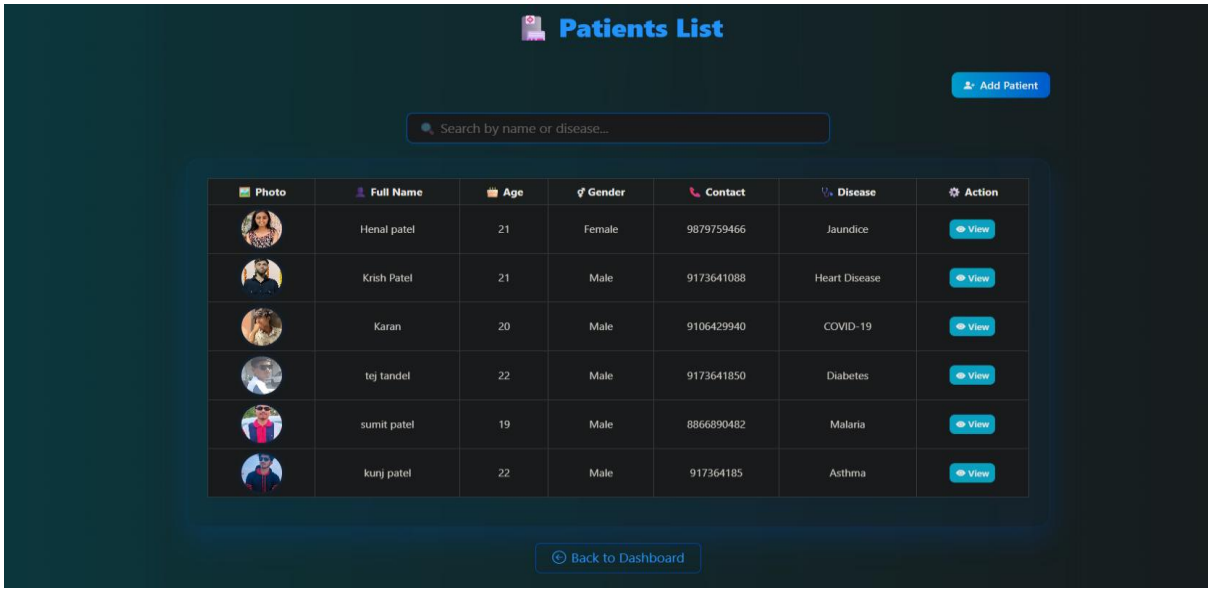


Figure 4.16 Patients Page

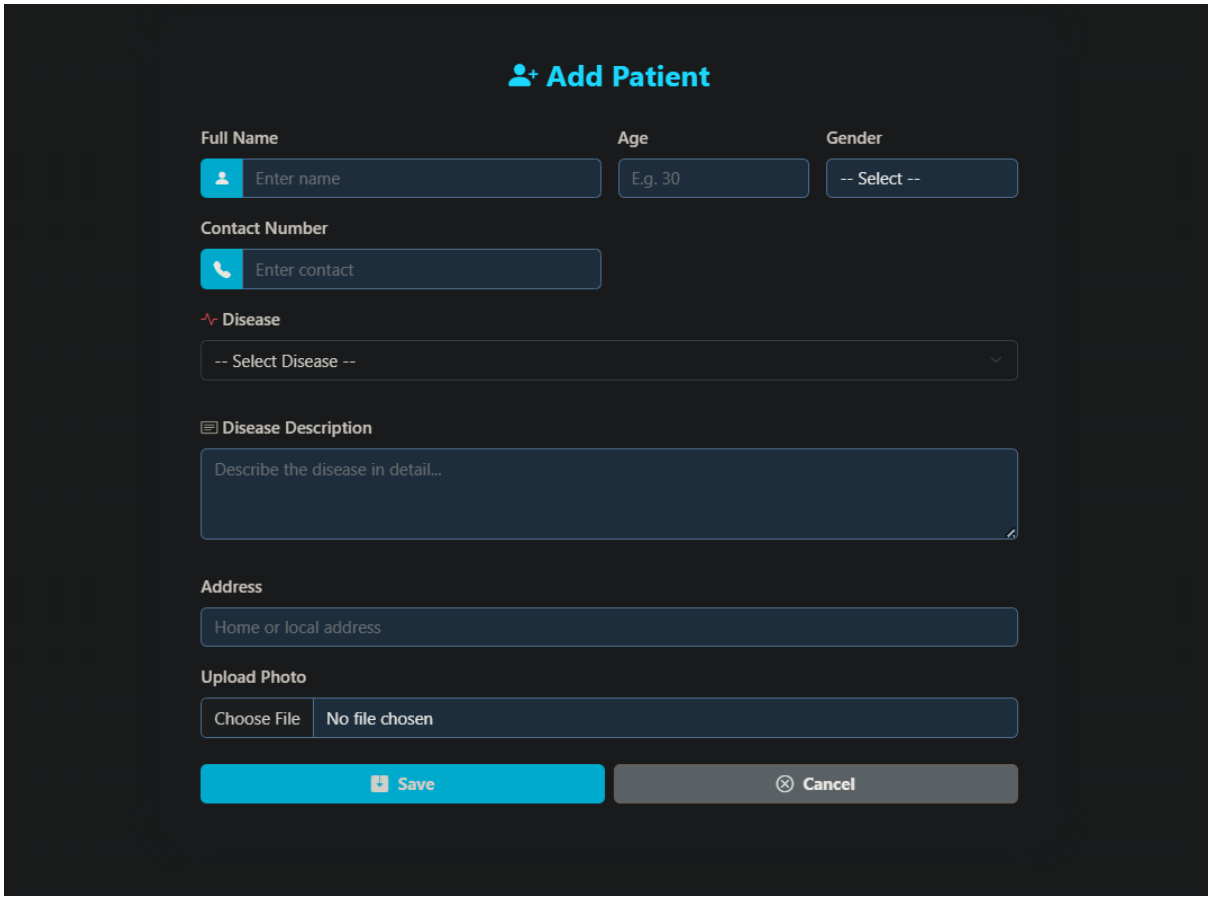


Figure 4.17 Add Patients Page

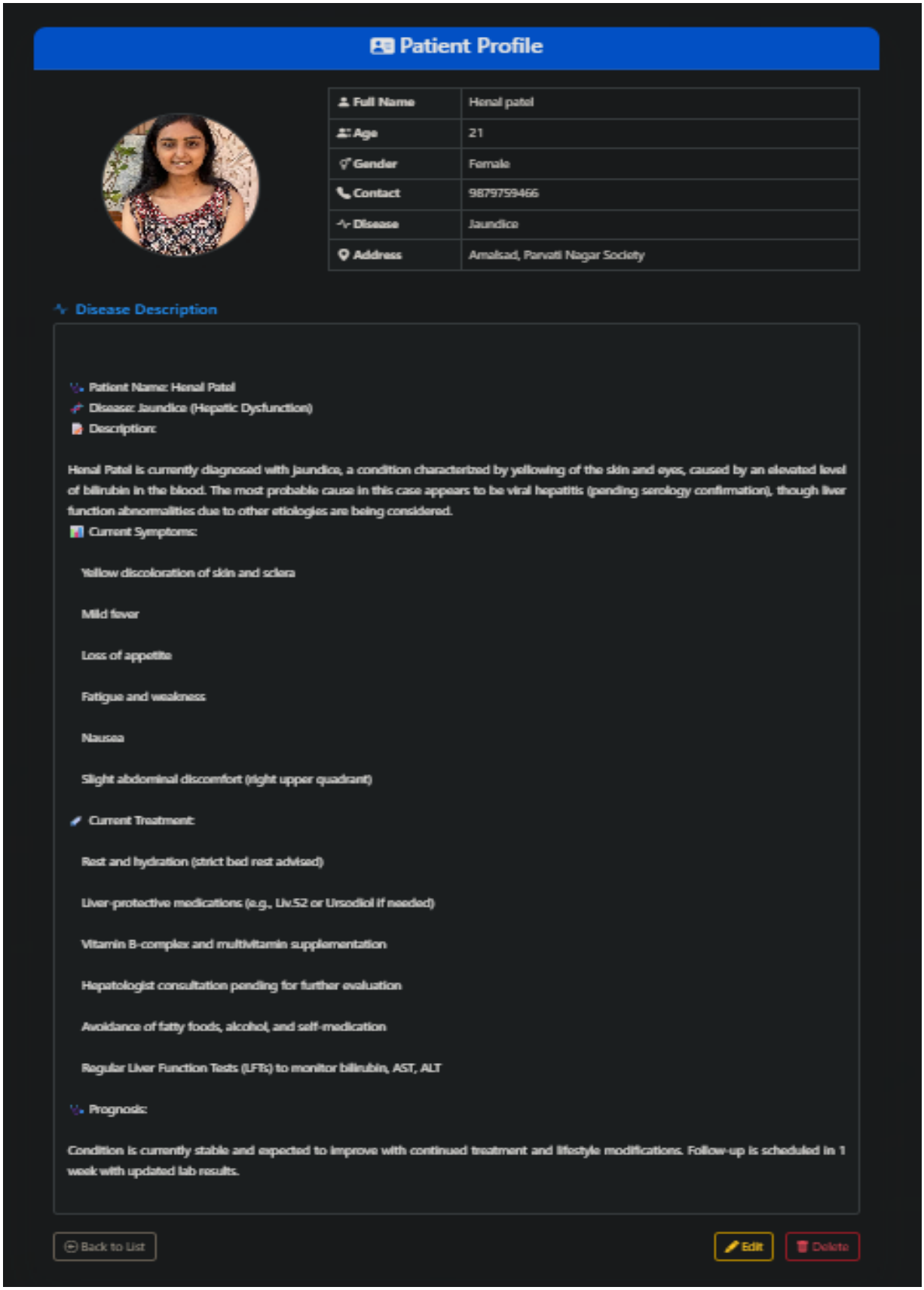


Figure 4.18 View Patients Profile Page

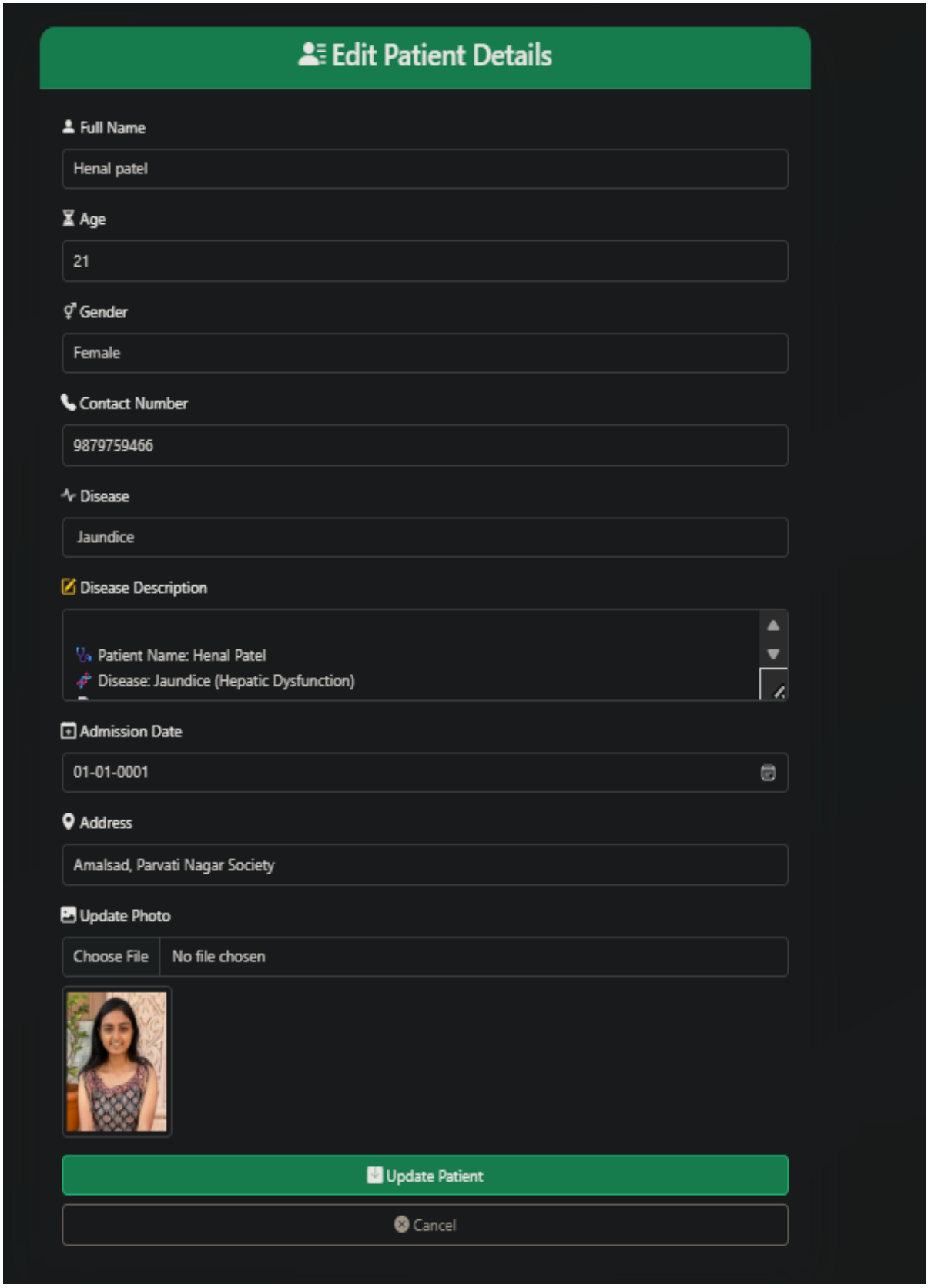


Figure 4.19 Edit Patients Profile Page

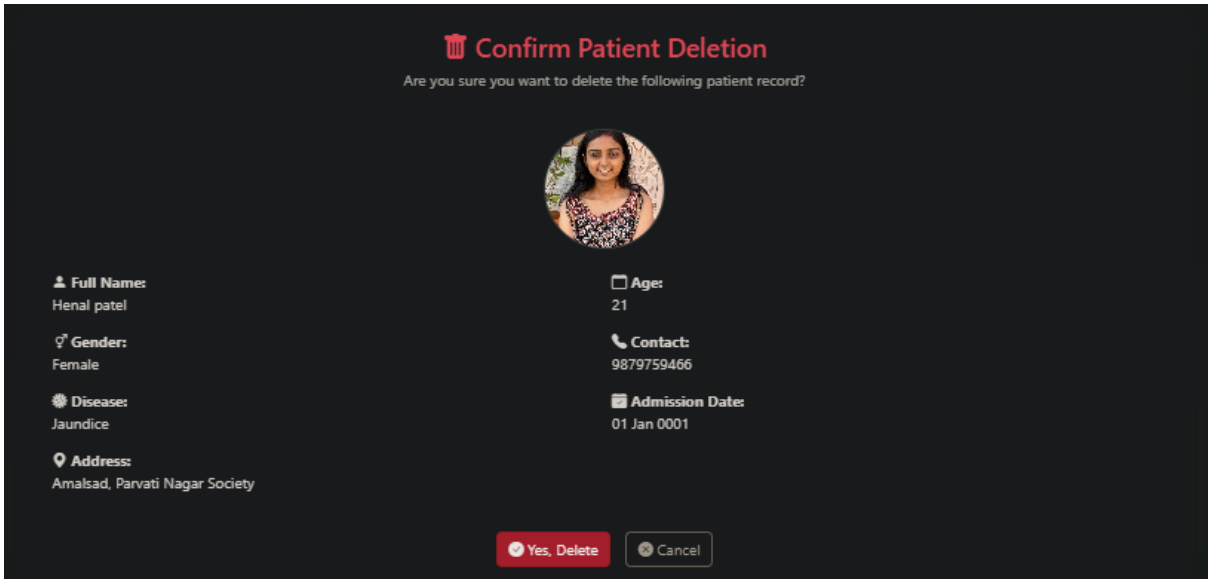


Figure 4.20 Delete Patients Profile Page

Room No	Type	Status	Assigned Patient	Action
1	ICU	Occupied	Krish Patel	Details
2	Private	Occupied	Henal patel	Details
3	Private	Available	None	Details
4	ICU	Available	None	Details
5	ICU	Available	None	Details
6	General	Occupied	kunj patel	Details
7	Private	Available	None	Details
8	Private	Occupied	tej tandel	Details
9	ICU	Occupied	sumit patel	Details
10	General	Available	None	Details
11	Private	Available	None	Details
12	General	Available	None	Details

Figure 4.21 Room Management Page

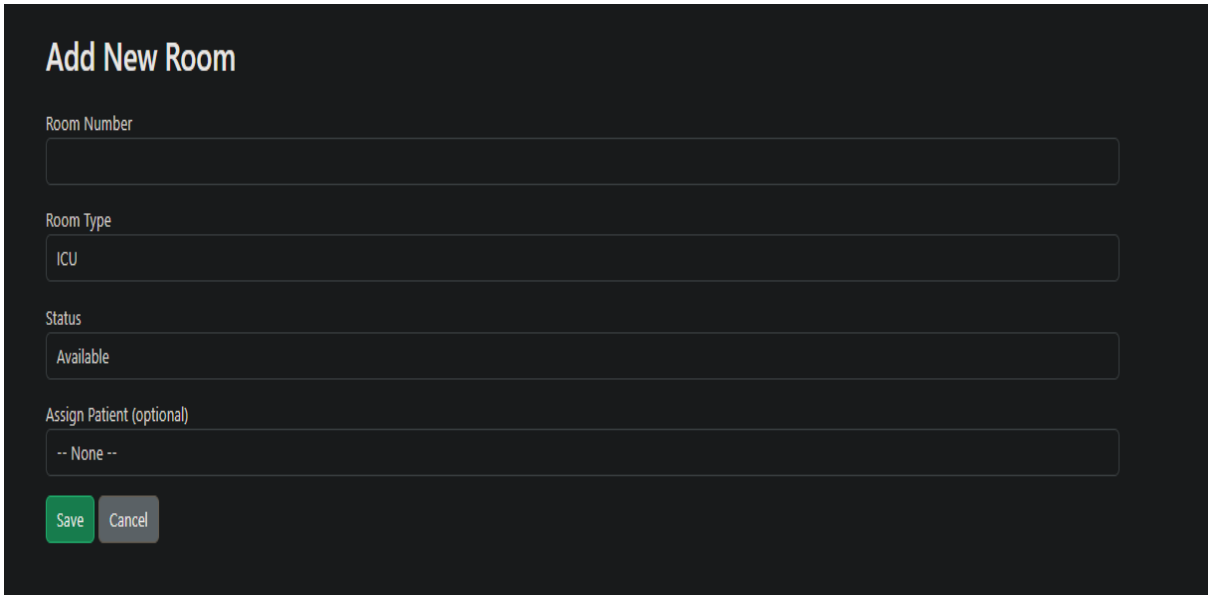


Figure 4.22 Add New Room Page

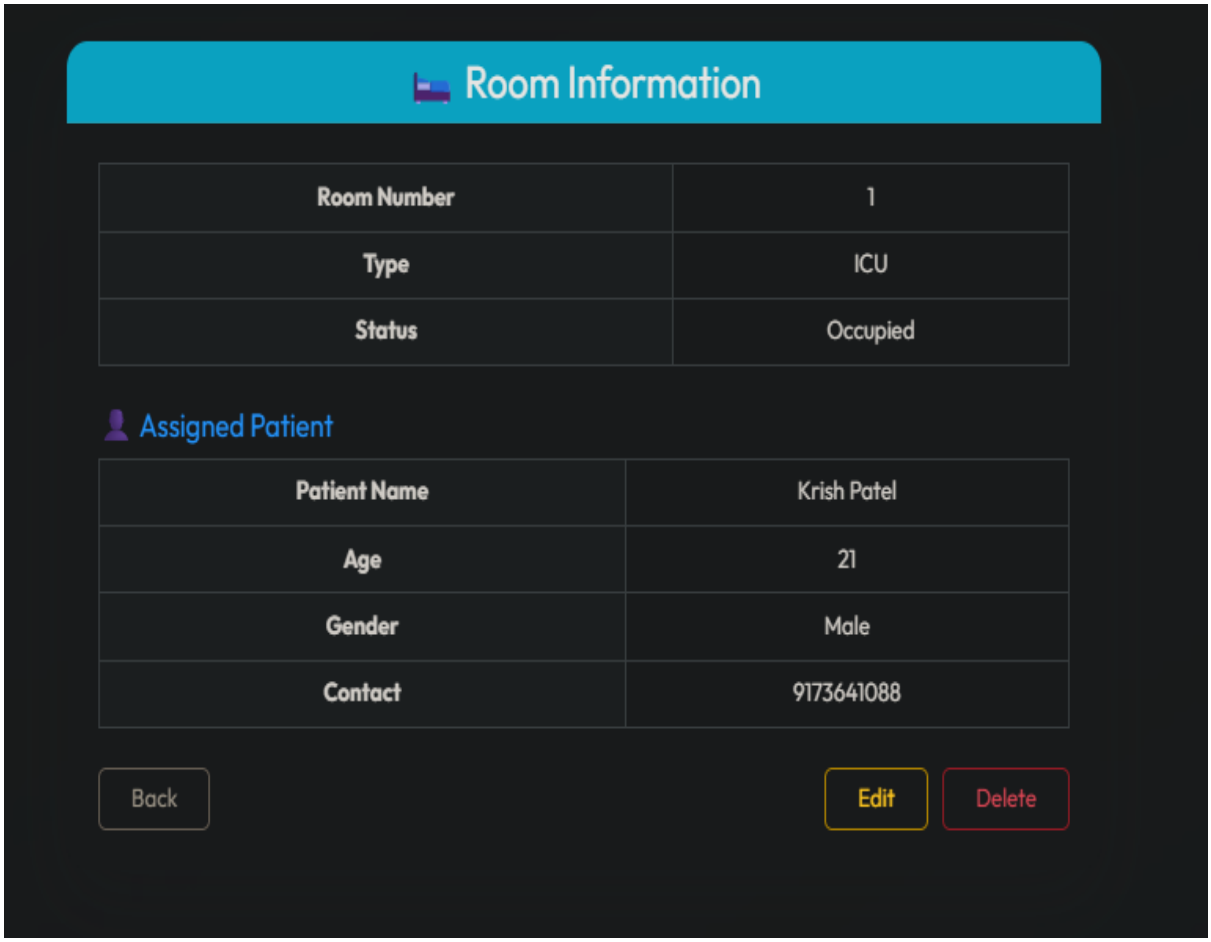


Figure 4.23 View Room Information Page

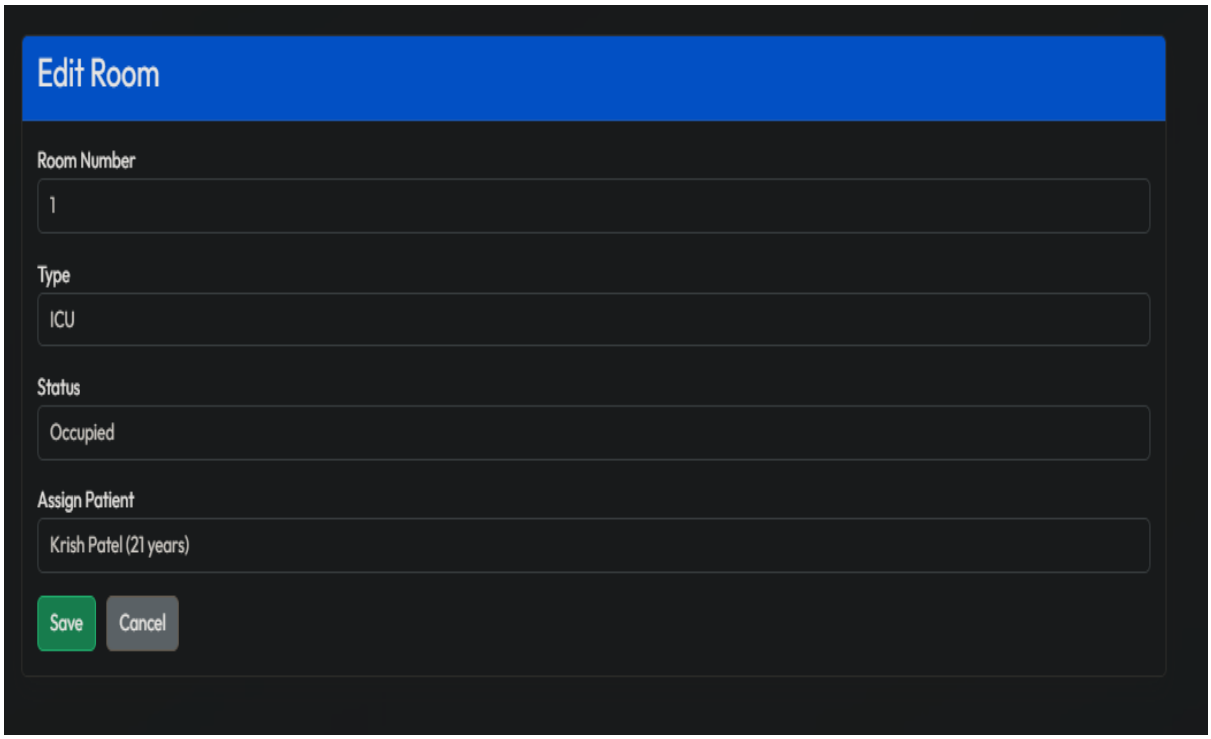


Figure 4.24 Edit Room Information Page

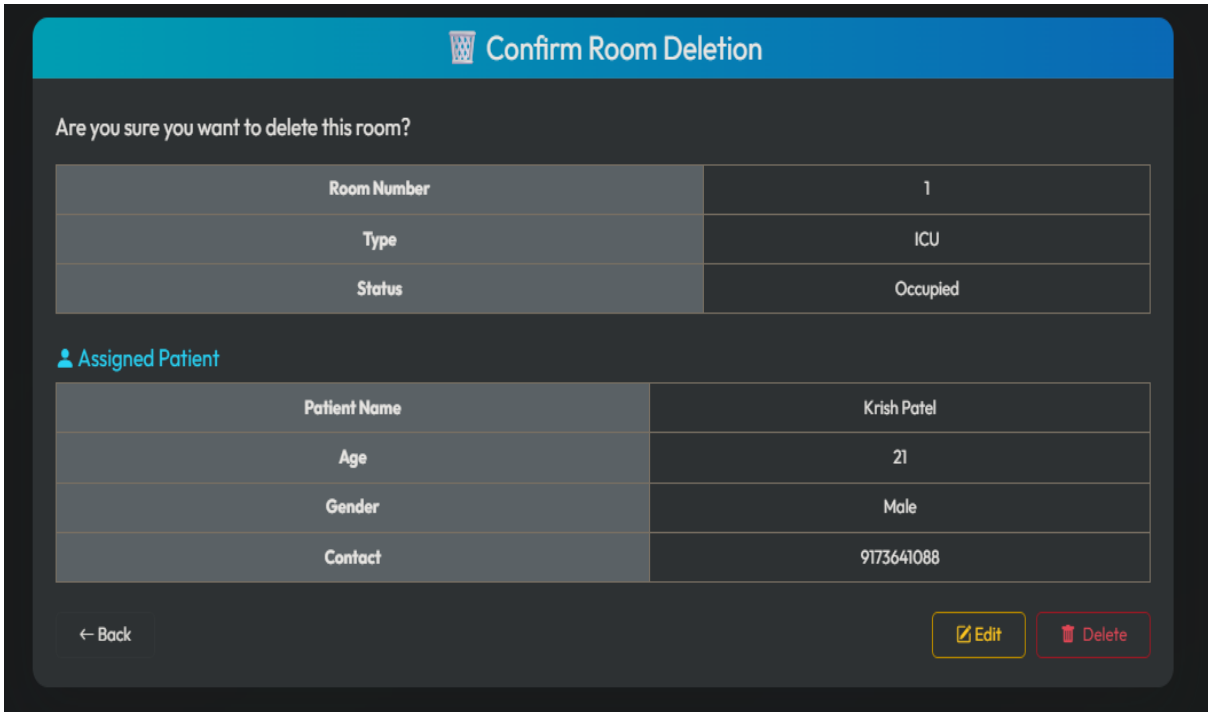


Figure 4.25 Delete Room Information Page

Figure 4.26 Billing Records Page

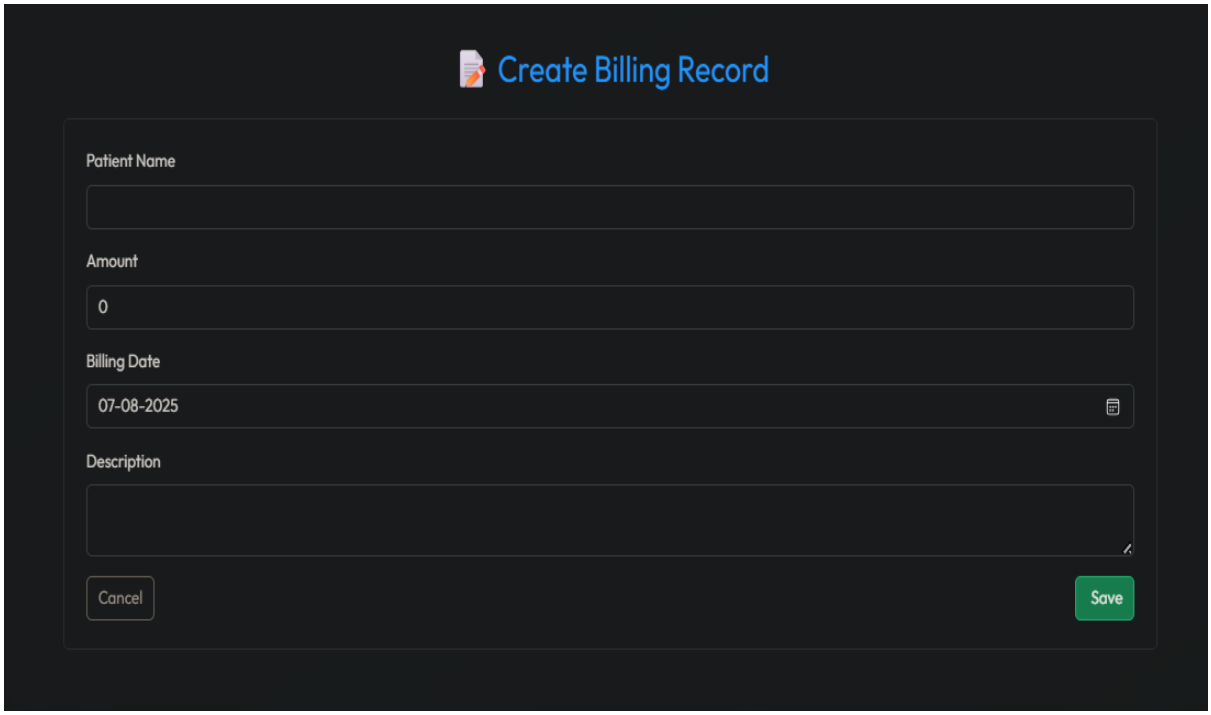


Figure 4.27 Create Billing Record Page

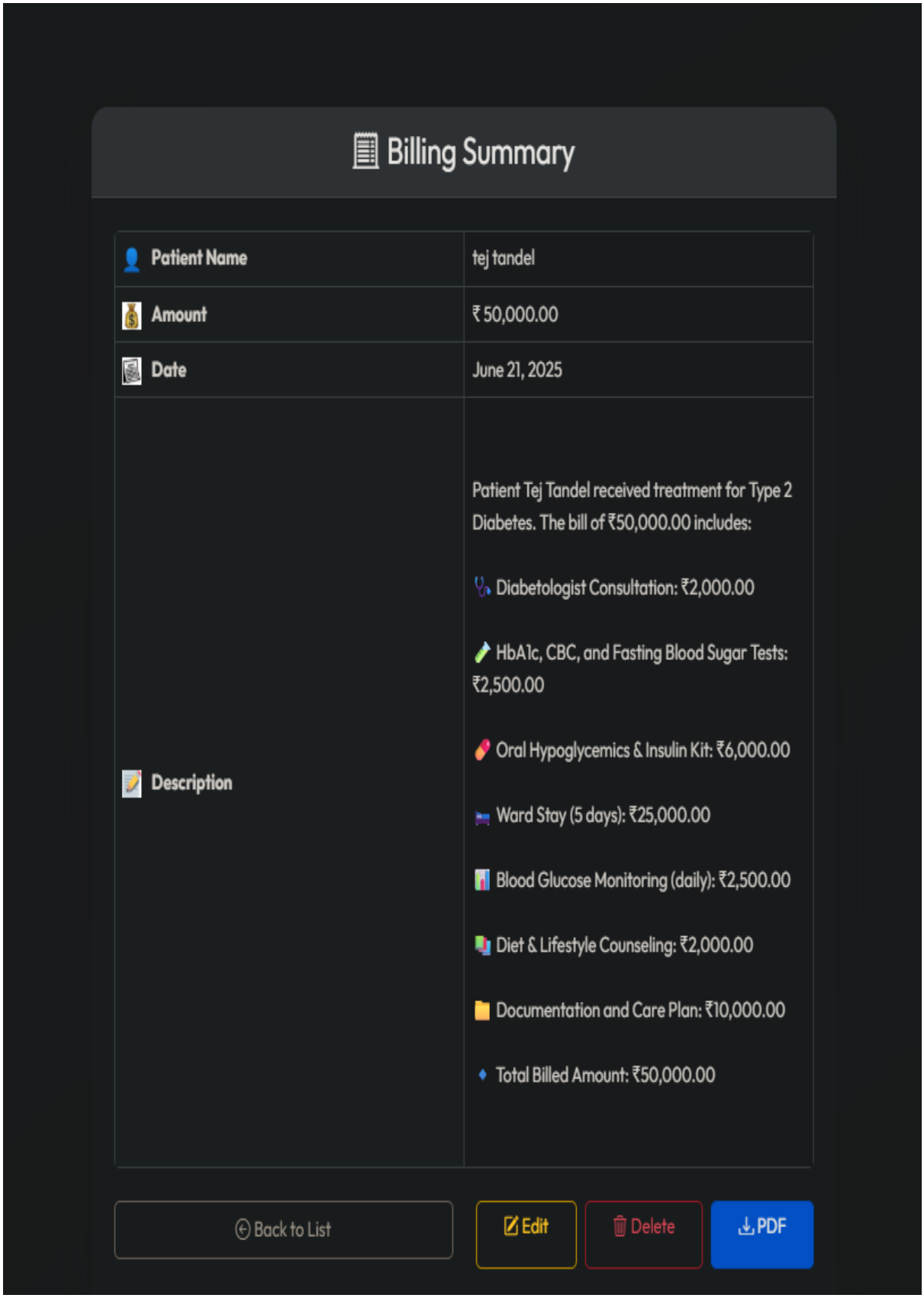
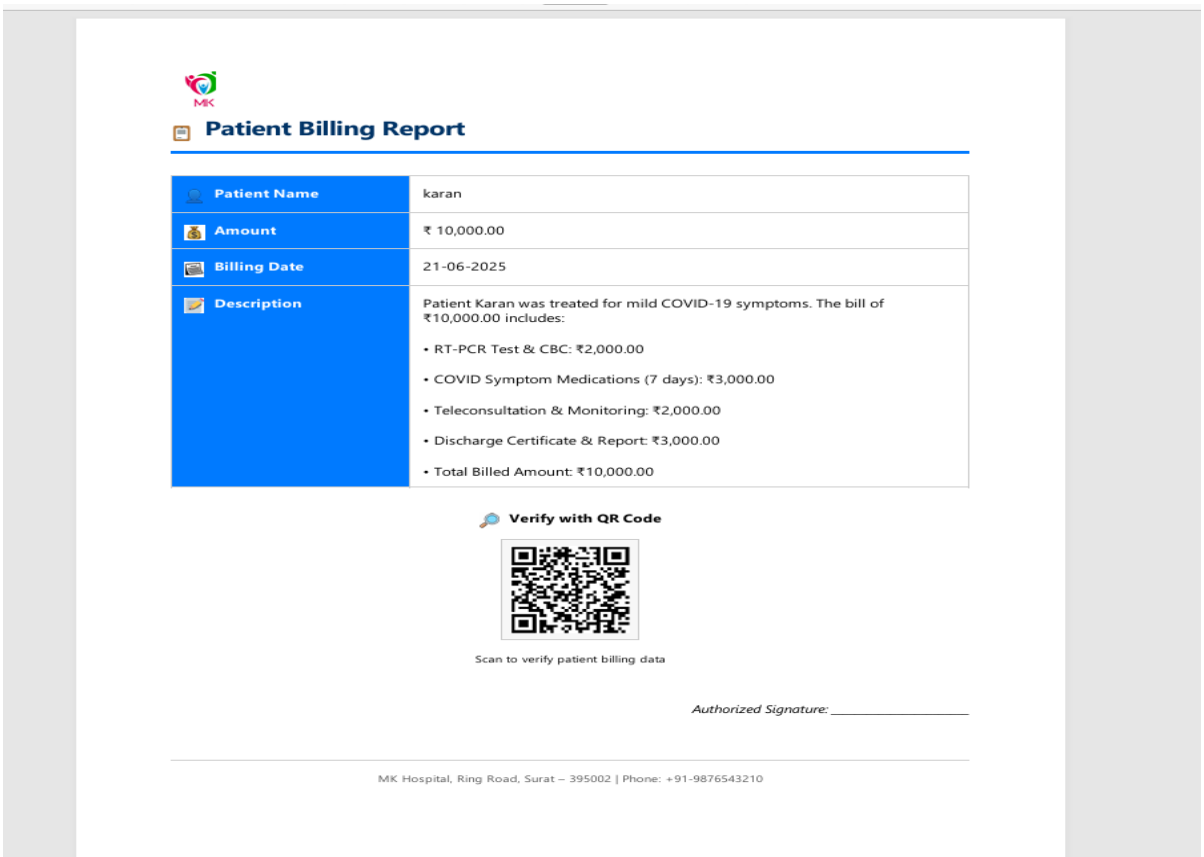


Figure 4.28 View Billing Summary Page

Figure 4.29 Edit Billing Record Page



MK

Patient Billing Report

Patient Name	karan
Amount	₹ 10,000.00
Billing Date	21-06-2025
Description	Patient Karan was treated for mild COVID-19 symptoms. The bill of ₹10,000.00 includes: • RT-PCR Test & CBC: ₹2,000.00 • COVID Symptom Medications (7 days): ₹3,000.00 • Teleconsultation & Monitoring: ₹2,000.00 • Discharge Certificate & Report: ₹3,000.00 • Total Billed Amount: ₹10,000.00

Verify with QR Code

Scan to verify patient billing data

Authorized Signature: _____

MK Hospital, Ring Road, Surat – 395002 | Phone: +91-9876543210

Figure 4.30 Patient Billing Report Page

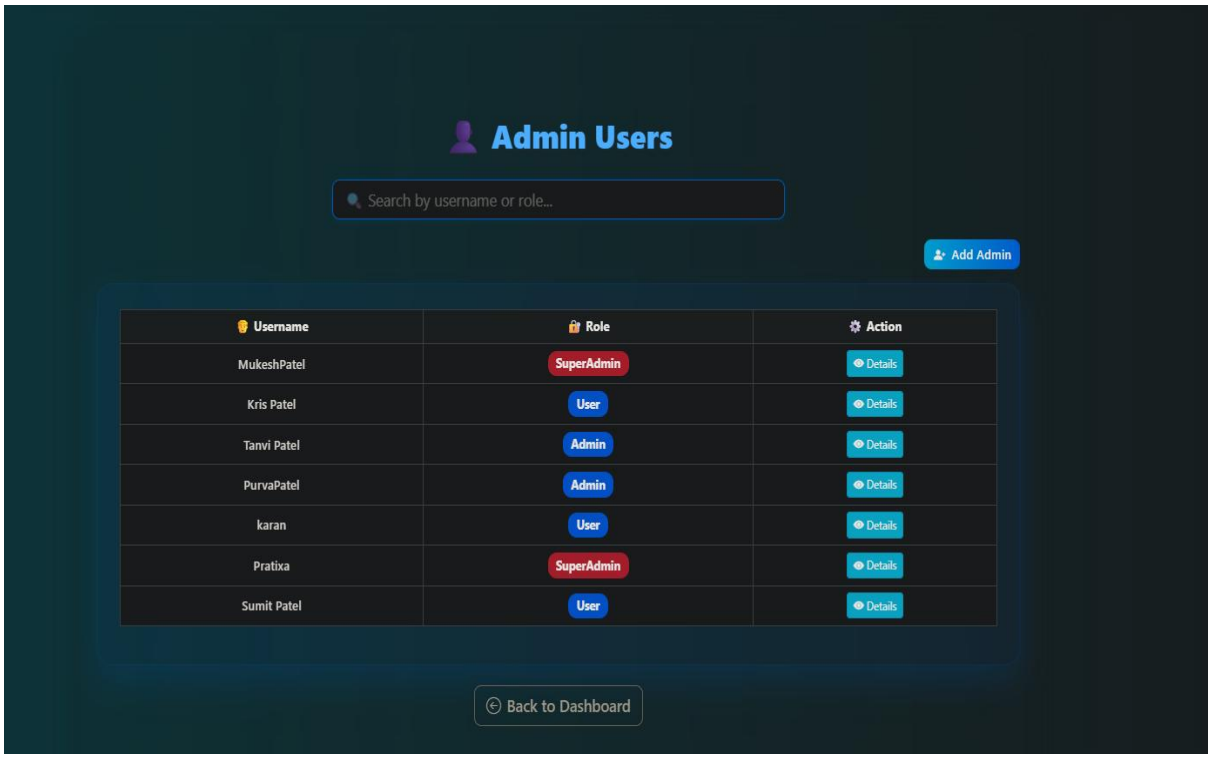


Figure 4.31 Admin User Page

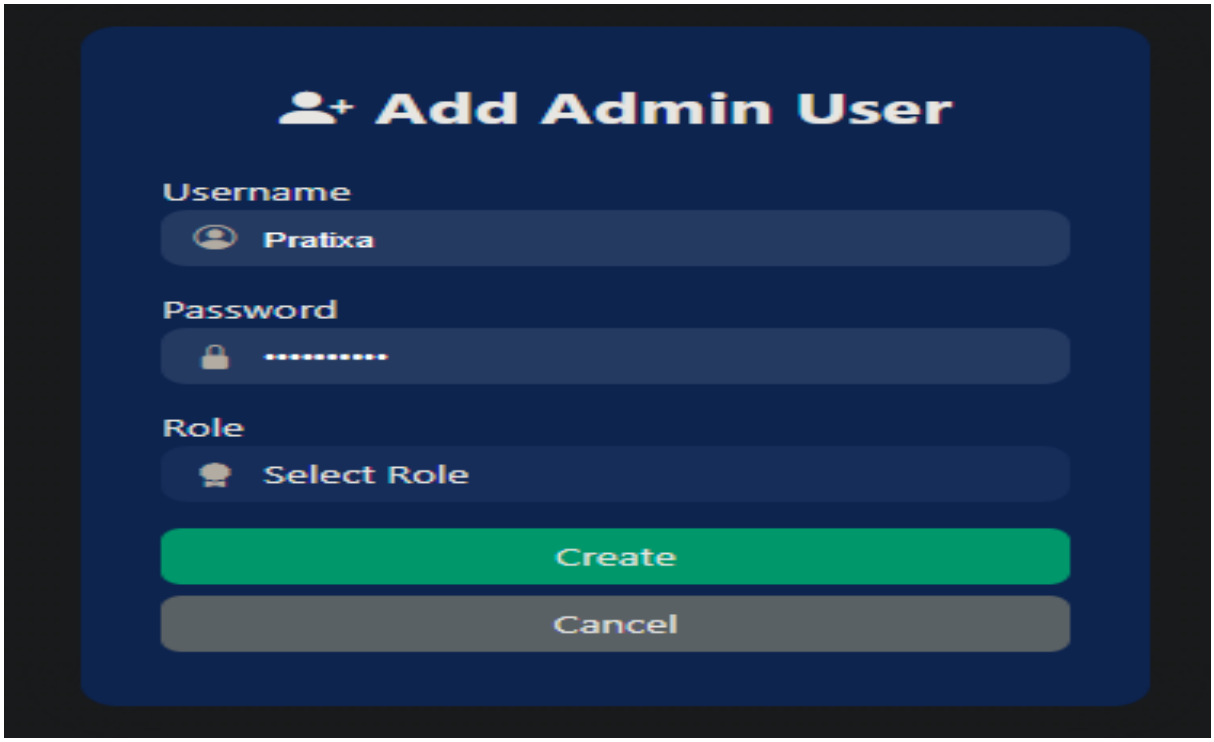


Figure 4.32 Add Admin User Page

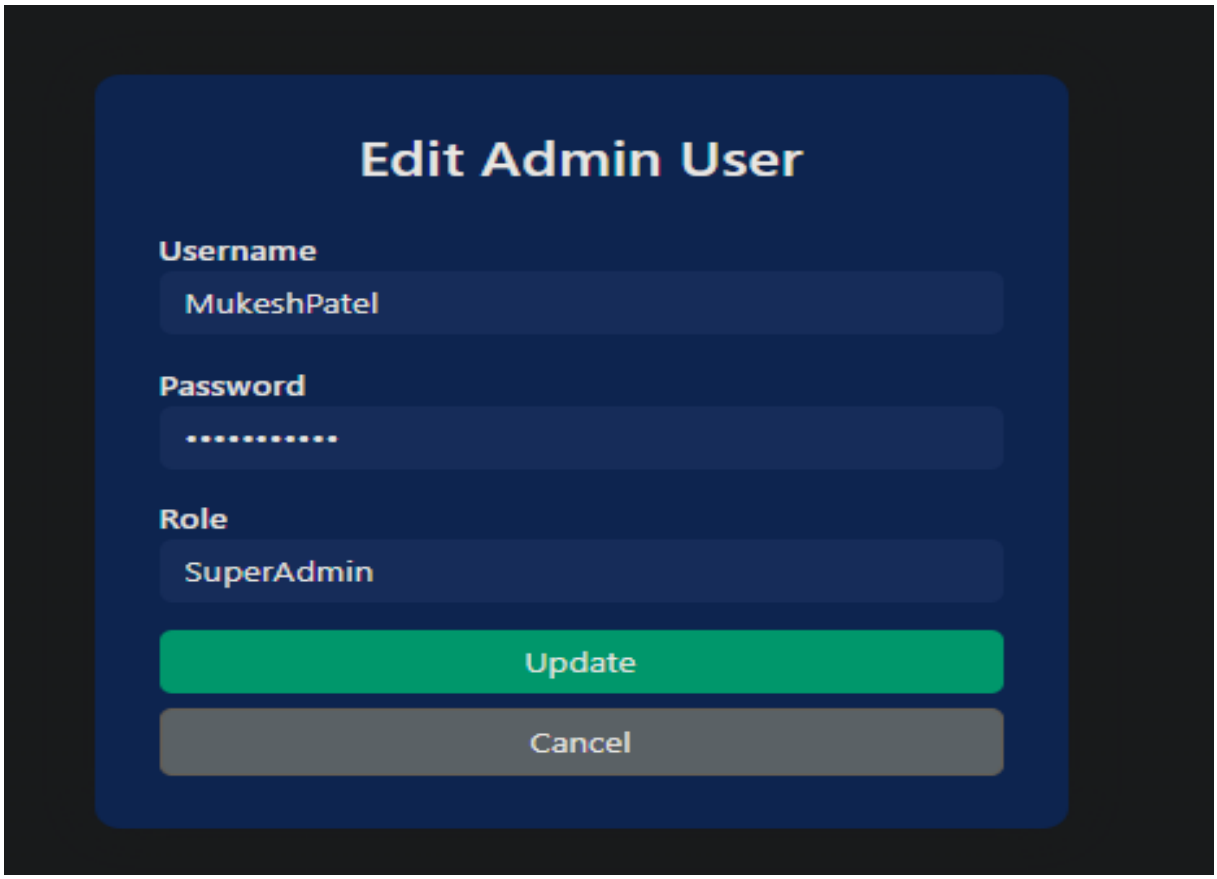


Figure 4.33 Edit Admin User Page

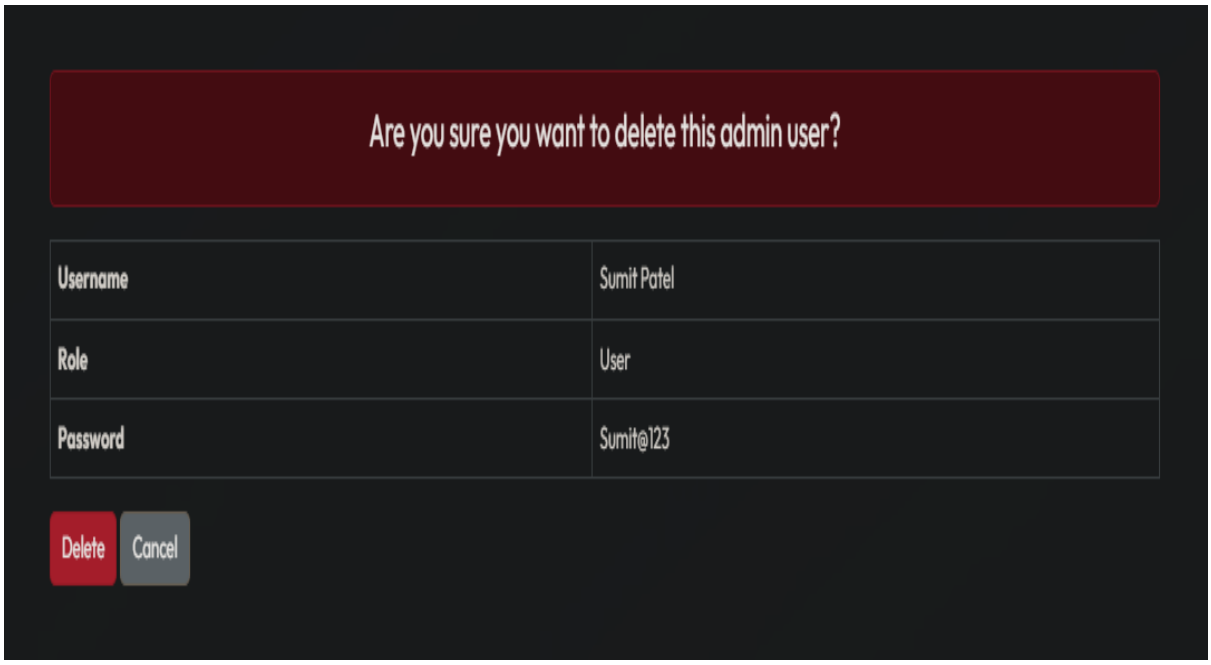


Figure 4.34 Delete Admin User Page

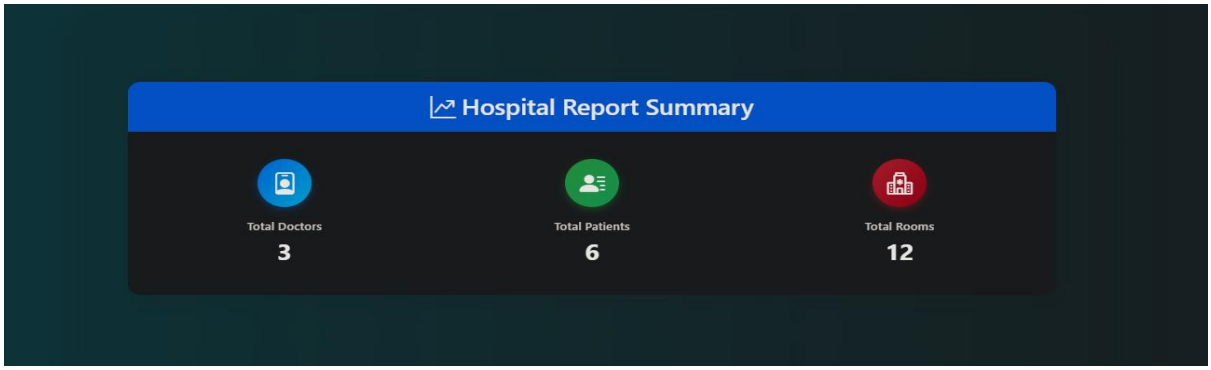


Figure 4.35 Hospital Report Summary page

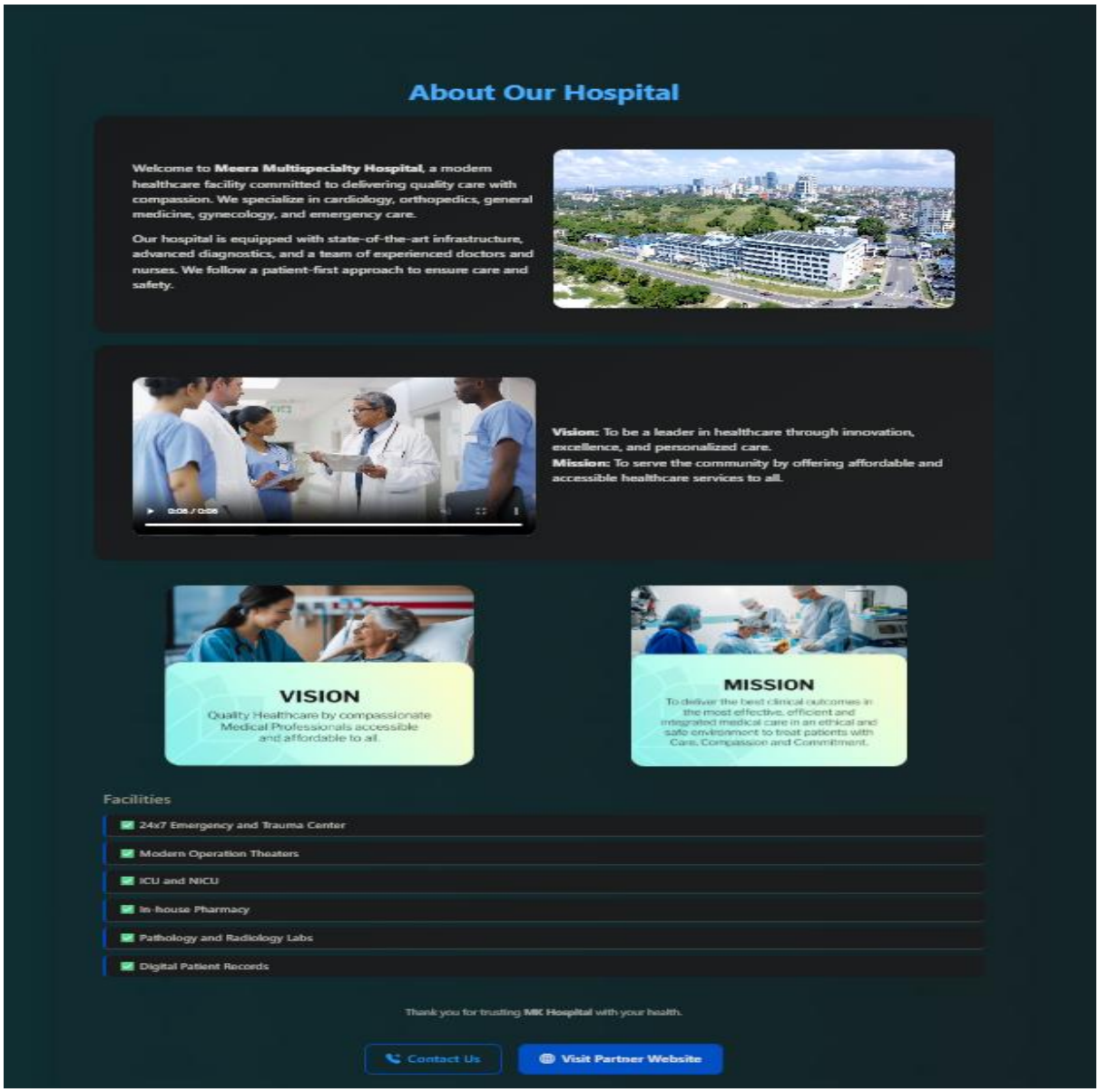


Figure 4.36 About Page

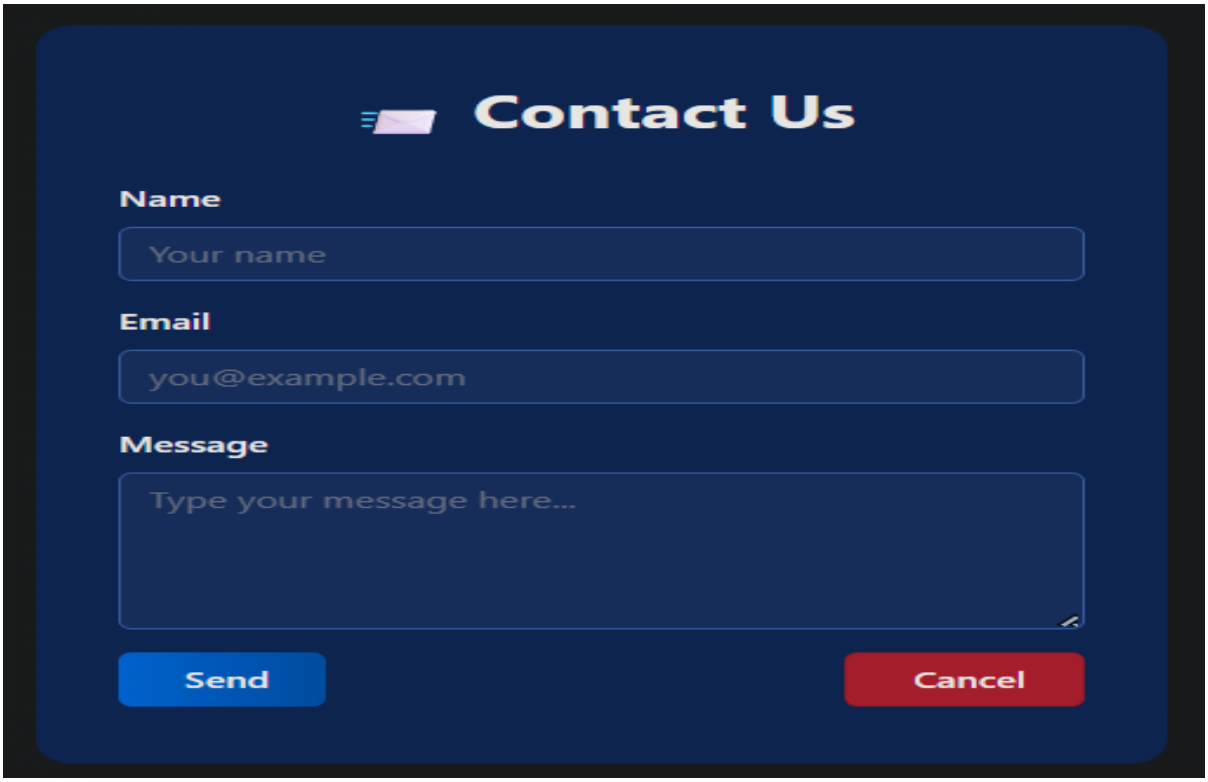
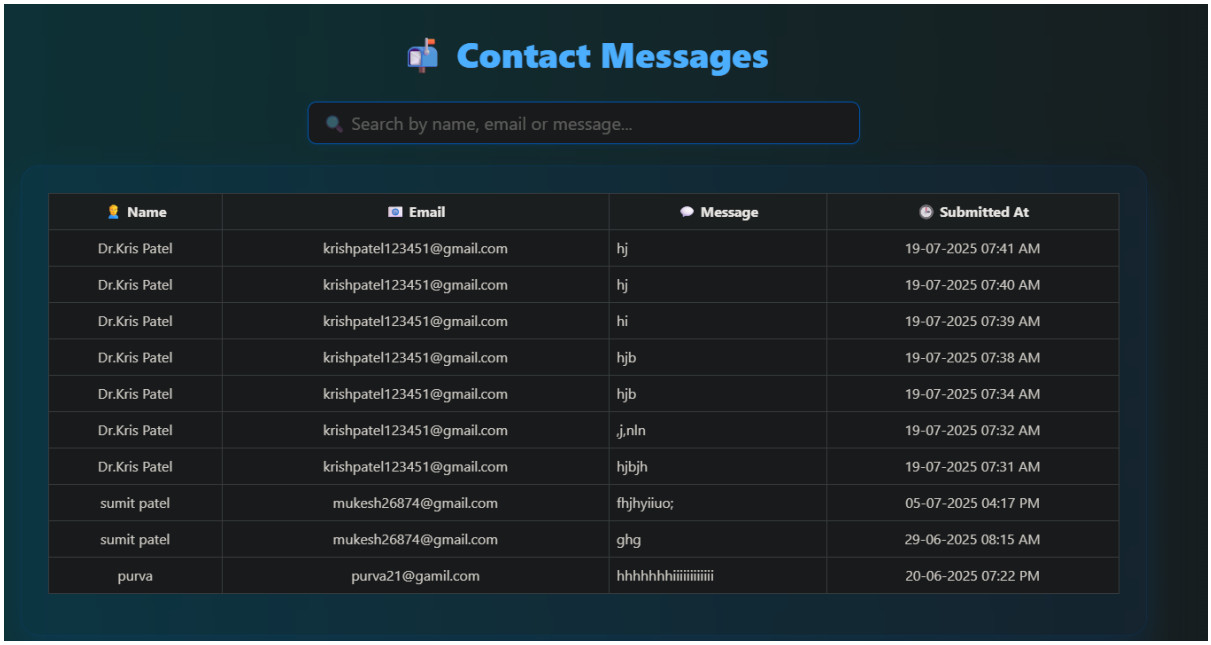


Figure 4.37 Contact Page



Name	Email	Message	Submitted At
Dr.Kris Patel	krishpatel123451@gmail.com	hj	19-07-2025 07:41 AM
Dr.Kris Patel	krishpatel123451@gmail.com	hj	19-07-2025 07:40 AM
Dr.Kris Patel	krishpatel123451@gmail.com	hi	19-07-2025 07:39 AM
Dr.Kris Patel	krishpatel123451@gmail.com	hjb	19-07-2025 07:38 AM
Dr.Kris Patel	krishpatel123451@gmail.com	hjb	19-07-2025 07:34 AM
Dr.Kris Patel	krishpatel123451@gmail.com	,j,nln	19-07-2025 07:32 AM
Dr.Kris Patel	krishpatel123451@gmail.com	hjbjh	19-07-2025 07:31 AM
sumit patel	mukesh26874@gmail.com	fhjhyiiuo;	05-07-2025 04:17 PM
sumit patel	mukesh26874@gmail.com	ghg	29-06-2025 08:15 AM
purva	purva21@gamil.com	hhhhhhhiiiiiiiiiii	20-06-2025 07:22 PM

Figure 4.38 Contact Messages Page

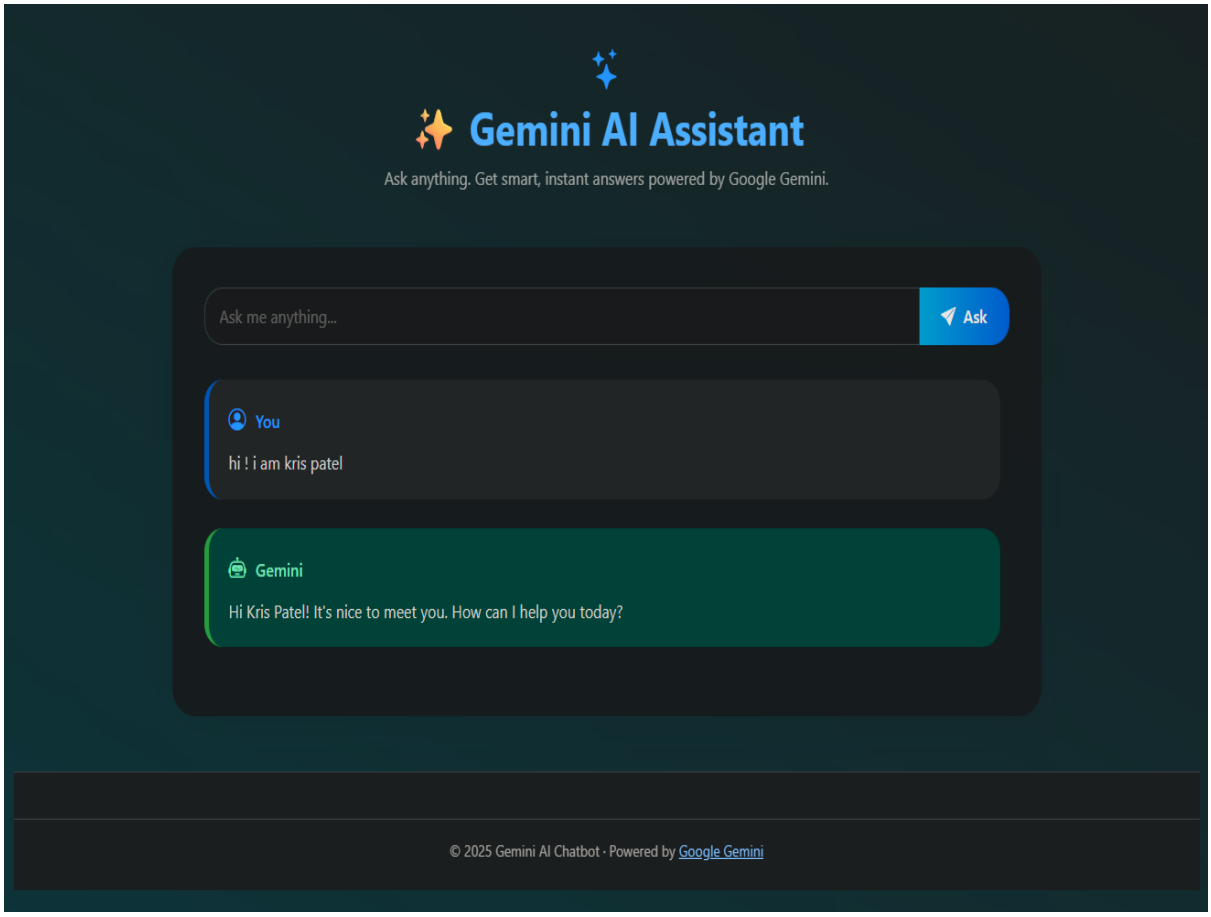


Figure 4.39 Gemini AI Assistant Page

CONCLUSION AND FUTURE SCOPE

Conclusion

The Hospital Management System (HMS) developed using ASP.NET Core MVC and SQL Server efficiently manages various hospital operations such as patient registration, doctor management, appointment scheduling, and billing. By implementing best practices in database design and MVC architecture, the system ensures security, scalability, and smooth data handling, resulting in improved workflow and reduced manual errors.

Furthermore, the project has enhanced understanding of full-stack .NET development and demonstrated how software solutions can significantly improve healthcare management. It not only streamlines hospital operations but also enhances patient experience, serving as a solid foundation for developing more advanced healthcare applications in the future.

Future Scope

In the future, the Hospital Management System can be enhanced with advanced features such as Electronic Health Record (EHR) integration, telemedicine, and mobile application support to make healthcare more accessible and connected. The inclusion of secure online payment gateways will also make billing more convenient for patients.

Additionally, the system can incorporate analytics dashboards, multilingual support, automated notifications, and biometric or RFID-based authentication for improved usability, data security, and efficiency. These enhancements will make the HMS more intelligent, user-friendly, and adaptable to the evolving needs of the healthcare industry.

REFERENCES

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https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://hospitalmanagementsystem.org/&ved=2ahUKEwiIx6W98PyOAxXjV2wGHfloJhsQFnoECCwQAQ&usg=AOvVaw0-NJfXkw1kCiUhA9po-amH